

Self-powered binomial sums

Corrected Poisson endpoint expansions and the optimal binomial law of the critical inactive count, for OEIS A360592, A360479 and A360747

Part I: corrected Poisson endpoint expansions

Part II: optimal positive binomial approximation for the critical inactive count

Two research reports of one external research session
(bundle Reports 230 and 235), bound into one ProveIt report

Both Parts: 5 October 2026 merged: 7 October 2026

Abstract

Two manuscripts on one family of exact finite sums, bound into one report: $a_p(n) = \sum_{0 \leq k \leq n/(p+1)} (n - pk)^{pk} \binom{n-pk}{k}$, which for $p = 1, 2, 3$ are OEIS A360592, A360479 and A360747. Part I (Report 230, the base) proves every fixed algebraic order for every fixed $p \geq 1$ from an exact residue-conditioned Poisson representation: a small deformation for $p \geq 2$ with explicit rational generators, and at the critical endpoint $p = 1$ a centered resummation with half-power corrections only and first correction $c/4 + 11c^3/8$, $c = \sqrt{e/2}$; it identifies the first corrections of the three OEIS refinements as wrong (an extra $1/(8c_p)$ at $p = 1, 2$, a nonzero $n^{-1/4}$ term at $p = 3$ whose true value is zero) while their leading equivalents stand; it proves, for $p \geq 2$, weighted density expansions, moment refinements and sharp total-variation constants for the inactive-vertex count against two conditioned Poisson references, and controlled Newton and Lagrange inverses with a two-ceiling threshold envelope. Part II (Report 235) answers Part I's question on the critical marked law: for $p = 1$ the globally optimal parity-conditioned binomial approximation of the inactive count has total variation $\sim \frac{5}{8} \log 2 \sqrt{2/\pi} \mu^{3/2}/n^2$, attained by an explicit law, with every asymptotically optimal parameter sequence characterized. The write quotes the three OEIS entries with their revision numbers, checks the source's transcription of their formulas exactly and the refutation against the exact sums, recomputes every coefficient, polynomial identity and printed decimal of both Parts with its own code, compares the inverses with the repository's transseries volume, and gathers the open questions. The report is unrefereed, and nothing in it is formalized.

[write] **Status and trust boundary, added 7 October 2026, batch 113.** This report binds bundle Reports 230 and 235 of an external AI-assisted research session into the ProveIt research report **a360592-self-powered-binomial-sums**; the Guide below gives the provenance, the sources as the write read them, what was checked, the relation to the repository, the collected non-claims and the reading conventions. Report 230's title block reads "Report 230" (PDF author "Research report"), Report 235's "Report 235" (PDF author "Report 235"): they name no person or tool. *What rests on what.* Every theorem has a conventional proof in the text: exact algebra, Poisson and binomial factorial-moment identities, root-of-unity filters, Chernoff and Cauchy estimates, Taylor's theorem with explicit weighted remainders, Charlier and Hermite polynomials, Lagrange inversion, the Lambert function, log-concavity and an exact characteristic function Jacobian. No proof uses a computation; the exact checks corroborate finite identities and every decimal is an uncertified diagnostic. The write adds Remarks 6.7 and 7.1 and dated notes, and prints three

short proofs of Part II only once (in Part I, see the Guide); no statement or number of either source was changed.

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Guide to this report

The two Parts are one chain. Part I is bundle Report 230 (the base of the merge): it treats every fixed $p \geq 1$ and leaves the marked law at $p = 1$ open (its Scope paragraph and Section 5 assume $p \geq 2$). Part II is bundle Report 235, which calls Report 230 its “scalar predecessor” and settles the critical marked law for the inactive count; “Report 230” in Part II is Part I. Part II reproves at $p = 1$ the inputs it uses. Its exact reduction (Lemma 10.1), defect expansion (Lemma 10.2), Poisson estimates (Lemma 10.3) and scalar expansion (Proposition 11.1, which extends Part I’s Theorem 4.1 to a compact range of the marking parameter) carry labels that later parts of Part II cite, so they are kept, each with a dated note naming its Part I counterpart. Three passages of Part II that only repeat a Part I proof, and carry no label, are printed once, in Part I: the proof of the global bound (10.3) (Part I’s proof of Lemma 4.2), the proof of Lemma 10.2 (Part I’s verification of (4.11)) and the direct check of the first scalar coefficient (Part I’s Section 4.5). A dated note stands at each place. Part II’s Corollary 11.4 is the case $p = 1$ of the two total-variation equivalents that Part I’s Theorem 5.1 proves for $p \geq 2$, with the same constants; it is new, not a repetition.

Numbering. Part I keeps every delivered number, including its Appendices A and B. Part II continues the section counter: Report 235’s Section k is Section $k + 8$ here, and since both texts number equations and statements within sections, its equation $(k.j)$ is equation $(k + 8.j)$ and its Theorem $k.j$ is Theorem $k + 8.j$ (its Theorem 1.1 is Theorem 9.1); its appendix keeps the letter A (“Part II’s Appendix A”). Labels carry **spb:ep:** (Part I, “endpoint”) and **spb:ml:** (Part II, “marked law”); the write’s own labels carry the Part prefix, or **spb:** for this Guide. The bibliography is merged at the end; Part II’s key **oeis** for A360592 is Part I’s **oeis1**, and every detail of both entries is kept.

[write] **Provenance, added 7 October 2026, batch 113.** Both manuscripts are dated 5 October 2026; they arrived in 60f54ea06 and were placed in a4186a946. *Part I:* Report 230, archive **Report230.zip** (630,758 bytes, 22 files, wrapper directory **Report230/**); text **article.tex** (100 lines) with nine section files (1677 lines), 30 pages as delivered, staged as **article.tex** and **sections/01_overview.tex–09_appendix.tex**. *Part II:* Report 235, archive **Report235.zip** (672,890 bytes, 28 files, wrapper **Report235/**); text **article.tex** (49 lines) with ten section files (848 lines), 23 pages as delivered, not staged; the write ships its first nine section files as **sections/235-binomial-01_results.tex–09_coupling.tex** and folds its tenth, the bibliography, into the merged one. Neither package names a person, tool or addressee, carries a “prepared for private review” line, an e-mail address or personal data, or records a ProveIt commit (so no pin is recorded). The programs are shipped under **code/** and the requirements and recorded outputs under **data/**, with the prefixes **230-endpoint-** and **235-binomial-**; the delivered root **build.py** of each is **code/2NN-**-build.py**; the source notes and guides are at the top level (**230-endpoint-SOURCES.md**, **230-endpoint-code-README.md**, **235-binomial-SOURCES.md**, **235-binomial-COMPUTATION.md**). Report 230’s delivered **README.md** was staged as the report **README** and is replaced by it. Not shipped, and retrievable from the arrival commit: the two delivered PDFs; Report 235’s **article.tex**, section files and **README.md** (its text is Part II of this file); and the two pure checksum manifests **MANIFEST.sha256** (21 and 27 entries), both verified at the write. *Where the merge had to choose:* the base (Report 230, which treats every p ; Report 235 is its $p = 1$ continuation and cites it); the order (the chain’s order); keeping Part II’s restatements, each with a note, because later labels depend on them, and printing three unlabelled repeated proofs once; one merged bibliography; the preamble (the union of the two, which agree where they overlap; Report 235’s global list spacing is not used, and the running heads are neutral). The write prefixed the 229 delivered labels (127 and 102) and updated the references to them (Part I: 95 **\eqref**, 26 **\ref**; Part II: 79 **\eqref**, 40 **\ref**), and added this Guide, the Part headers, Remarks 6.7 and 7.1 and the dated notes. The added remarks are the

last statements of their sections and the added displays are unnumbered, so every delivered number is unchanged up to the stated shift (label tables of builds of the two delivered texts and of this one compared: 229 labels, 0 differences).

The sources, as the write read them (7 October 2026). OEIS A360592, A360479 and A360747 in the internal format with their b-files (Remark 7.1); the transseries volume `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex` (`p0:def:model, p0:thm:lambert-core, p0:def:three-inverses, p0:thm:staircase, plt:def:lw-monomial-log-datum, plt:thm:lw-template`). Not read by the write: the cited literature, in particular Kotešovec’s 2013 note, the DLMF and Corless et al. entries, and the binomial-approximation papers of Part II (Soon; Barbour–Xia; Čekanavičius–Peköz–Röllin–Shwartz; Barbour–Kowalski–Nikeghbali; Chhaibi–Delbaen–Méliot–Nikeghbali; Méliot–Nikeghbali–Visentin; Saumard–Wellner; Dümbgen–Mösching; Ehm). The Parts’ accounts of what they inspected (Section 7.4 and the subsection after it, and Section 15) are reported, not confirmed.

What was checked at intake. The batch-113 dossier read both manuscripts, found no demonstrably wrong claim inside them, confirmed from the exact sums, independently of the package, that the first corrections of the OEIS refinements are wrong while the leading equivalents are right ($p = 1$ to $n = 32001$, $p = 2$ to $n = 240000$), and reran the delivered suites on copies (all reproduce; Report 235’s guard tests reject Windows backslash paths, a Windows-only failure). *What the write checked.* It read both manuscripts in full and found no error. With its own code (Python 3.14.4, exact integers; SymPy 1.14.0; mpmath 1.3.0 at 50–110 digits), not importing the delivered programs, it

- computed $a_p(n)$ from (1.1) and found it equal to all terms of the three OEIS b-files (761, 636 and 601 terms), to the coefficients of the formal series (2.1) through degree 40, and to Proposition 2.1 for $p \leq 5$, $n < 80$;
- checked (3.6) against the Taylor coefficients of $\log G_r(z)$ for $r \leq 8$, $j \leq 5$, $p = 1, 2, 3$; the general- p forms (3.14)–(3.15), $b = A - q/2$, (3.3), and the critical list $D_1, \dots, D_5$;
- regenerated, by its own implementation of the recipes, every printed coefficient: (3.16), $C_{2,4}$, (3.17), $C_{3,5}$, $C_{3,6}$, the vanishing of $C_{p,1}, \dots, C_{p,p-2}$ for $p = 4, 5$ (the first nonzero coefficients are $-81c^2/10$ and $-151c^2/12$), (4.2)–(4.5), L_2 for $p = 2, 3$ and (5.21), the Charlier, falling-factorial and tilt identities (5.17), (5.15); and in Part II (11.5), (11.6), (9.5), (9.6), the factorial cumulants of Section 15.1, (12.12), (12.13) (the $O(1)$ term of M_0 is $7/6 - 8\theta^2/9$ before rounding), (12.3)–(12.6) (also against the series of $H_{M,\nu}$), the normalizer expansion and (12.14), the two log-density identities (12.16) and (12.17) through weight five with zero residual, their difference (12.18), (12.19) and the orthogonality of C_3 , the two hierarchy identities of Section 12.3, the signed moment identities (14.2) for $M \leq 8$, and the Gaussian constants (12.20), $\mathbb{E}|Z^2 - 1| = 4\phi(1)$, the median and $\mathbb{E}|W - m| = m$ of Lemma 13.2 (with J evaluated numerically at and around its minimizer);
- confirmed the critical coefficients numerically against the exact sums: with its own $C_{1,5}$ and $C_{1,6}$, $(a_1(n)/M_1(n) - \sum_{j \leq 6} C_{1,j}t^j)/t^7$ is $-5.50, -10.74, -11.30, -11.35$ at $n = 10^4, 10^6, 10^8, 10^{10}$; the analogous order- t^{13} ratios for $p = 2, 3$ are stable (about $2.8 \cdot 10^9$ and $1.2 \cdot 10^{12}$);
- recomputed every printed decimal of both Parts (notes in Section 7.2 and the subsection after it, and in Section 16), with the exact laws taken from log-gamma values of the summands, not from the delivered adjacent-ratio recurrence;
- checked the source’s transcription of the three OEIS formulas symbolically and the refutation numerically (Remark 7.1).

It reran, on fresh copies of the shipped programs (prefixes removed), Report 230’s `selftest.py` (normal, -0 , and `--numerical`: “PASS: 89 checks (optimized Python)”, “PASS: 653 checks (ordinary Python)”, the recorded result) and `diagnostics.pysuite-{}-deep`, and Report 235’s `checks.py` (normal and -0) and

`numerics.py`: every output equals the shipped file. The guard suite and the PDF/ZIP builders were not run. Floating computations are observations, not certificates.

Relation to the repository. Before batch 113 no file of the repository named A360592, A360479 or A360747. Other reports of the collection use some of the same tools for other objects: Poisson–Charlier expansions (`a340021-distinguished-maximal-independent-sets`, `a138178-symmetric-packed-matrices`, `a189281-path-forest-expansions`) and a total-variation binomial approximation (`a182220-source-boundary`). No result is shared, so no reciprocal note is proposed. No Lean or Rocq file treats this family; no statement of this report is formalized, and its place in the collection confers no formal status. The inverses are compared with the transseries volume in Remark 6.7.

[write] **What is not claimed, collected from both Parts.** *Part I:* every asymptotic assertion keeps p and the order fixed; no secondary exponential sector, optimal truncation, growing- p limit, effective inverse onset, certified constant or critical marked-law theorem; the transcendental diagnostics are not interval-certified, and the $p = 3$ truncation through t^{12} is still about 66% wrong at $n = 10^4$; the OEIS comparison concerns snapshots without verified revision histories, and no first-priority or exhaustive-search claim is made; the classical Poisson, Charlier, root-filter, Taylor, Newton, Lagrange and Lambert machinery is credited. *Part II:* the infimum is over the parity-conditioned binomial family only; no exact finite- n minimizer, no optimality over all positive families, no effective error certificate, no claim for other p or a growing modulus; positive binomial fitting, Charlier corrections and Hermite absolute-moment limits are credited to the literature it names, and the median minimization is called elementary L^1 projection; no worldwide priority claim and no claim of external review; the coupling appendix is supplementary and makes no global approximation claim. *The write adds:* its checks are exact or floating computations; its transseries remark claims no novelty for any inversion.

Reading conventions (letters used in more than one sense, within or across the Parts, with the tempting false readings; no symbol was renamed).

Symbol	Meanings
a	$a_p(n)$ the sums (Part I); $a(n) = a_1(n)$ (Part II); $a \equiv n \pmod{q}$ the residue (Part I, Section 5) and $a = n \bmod 2$ (Part II); a residue in (2.7).
q	$q = p + 1$ (Part I). Part II: q the success probability of $\text{Bin}(M, q)$ (Sections 9 and 14) and a modulus in its Appendix A. <i>False reading:</i> Part II's q is never $p + 1$.
c, d	$c_p = \exp(p^2/q)q^{-1/q}$, $c = c_1 = \sqrt{e/2}$; in Part II c also varies over a compact set (marking). $d = 3e/8 = 3c^2/4$, d_p ; in Part II d a positive constant in exponential bounds and $d_0 = \nu_0 - v_0$; d a degree index.
M, N	$M_p(n)$, $M_p(x)$ the normalizations (Part I); M the binomial trial count, M_0 , M_s , $M(\beta)$ (Part II); M the accuracy exponent of Corollary 6.3. $N = n/q$ (Section 2); $N = N(Y)$ the Lambert core (Section 6); N the support length in Part II's Appendix A.
$B, \mathcal{B}$	$B = N^{pn/q}$ (Part I); $B_{d,k}^{\geq 2}$ partition counts (both); $B_{M,\nu,a}$ conditioned binomials and $B_j(r;\nu)$ binomial log coefficients (Part II); $\mathcal{B}_a$ the conditioned binomial family; B_n a competitor.
C	$C_{p,j}(c)$ the coefficients (Part I), $C_j(c) = C_{1,j}(c)$ (Part II); $\mathcal{C}_2(r;\mu)$ the Charlier polynomial (Part I) and $C_1, C_2, C_3(r;\mu)$ (Part II); $\mathcal{C}_n$ the central event. <i>False reading:</i> Part II's $\mathcal{C}_2(r;\mu)$ is Part I's $\mathcal{C}_2$, not the coefficient $C_2(c)$; the arguments distinguish them.
ν	$\nu_K = (K + 1)/q$ an exponent (Section 6); in Part II ν the unconditioned binomial mean, ν_0 , ν_1 , $\hat{\nu}$, $\nu(\alpha)$.
γ, θ	$\gamma_1, \gamma_2, \gamma_3$ the coefficients of D_2 (Part I); $\gamma = 5/8$ (Part II). $\theta = z d/dz$ (Part I); $\theta = t\mu$ (Part II), with $\Theta = cd/dc$ there.

Symbol	Meanings
α, β, κ	$\alpha = c_p, \beta = 1/q, \kappa = p/q$ (Section 6); $\kappa_q = 1 - \cos(2\pi/q)$; $\beta_0 = \log(\mu/\lambda)$ (both Parts); α, β the local mean and trial parameters and κ_h cumulants (Part II).
L, H	$L_j(z)$ the cumulant polynomials (5.25) (Part I) and $L_j(c)$ the logarithmic coefficients (11.6) (Part II); L_* ; L an order. $H_j(r)$ (Part I), $H_N(z)$ (Lagrange), $H_{M,\nu}(r)$ the binomial log ratio and H_1, H_2, H_3 Hermite polynomials (Part II).
U, V, T	$U_j(z)$ the Poisson transforms (Part I) and $U = t(X - \lambda)$ the centered variable (both); V_K the weighted exponents (both; different centerings), $V_n, V(r)$. T_j Touchard polynomials, $T_p(y)$ the threshold (Part I), T_n a normalizer (Part II).
W, Z	W the Lambert function; $W_n(r)$ the weights and $\mathcal{W}_n$ the window (Part II); $W = Z^2$ in Lemma 13.2. Z a standard normal; $Z = Y + \log q + d_p$ (Section 6); $Z_n, Z_{K,a}$ normalizers.
ϵ, ρ, h	$\epsilon_n(r)$ a remainder and ϵ a radius (Part I); $\epsilon_n = \mu^{3/2}/n^2$ the scale (Part II). ρ_n, ρ_r (Part I), $\rho = 1 - 2q $ (Part II). $h = 2 \log 2 - 1$, $h_{K,J}$ (Part I); $h_0, h_{\alpha,\beta}$ densities (Part II).
J, K	J Newton steps (Part I), $J(\alpha, \beta)$ the Gaussian criterion and J_∞ the Jacobian (Part II). K an order; K_n active vertices, $K_2; K_a$ quotients in Part II's Appendix A.

Part I

Corrected Poisson endpoint expansions

Bundle Report 230, archive `Report230.zip`, delivered as `article.tex` with nine section files (30 pages as delivered), the base of the merge. Delivered title block “Corrected Poisson Endpoint Expansions / Critical resummation, inactive vertices, and controlled index inverses for a family of self powered binomial sums / Report 230 / 5 October 2026”. Printed as delivered, with `[write]` remarks and notes; its section, statement, equation and table numbers and its appendix letters are those of the delivery, and its labels carry `spb:ep:..`. Its bibliography is merged into the one at the end of the report.

Abstract of this Part (as delivered). For the exact sums

$$a_p(n) = \sum_{0 \leq k \leq n/(p+1)} (n - pk)^{pk} \binom{n - pk}{k},$$

we prove every fixed algebraic order for each fixed integer $p \geq 1$. An exact residue-conditioned Poisson representation gives a small deformation when $p \geq 2$ and a critical, nonvanishing deformation when $p = 1$. The latter requires a centered resummation and has half-power corrections, with no odd quarter powers. We calculate corrected coefficients for OEIS A360592, A360479, and A360747; their observed leading equivalents remain valid. For $p \geq 2$ we prove weighted density expansions, mean and variance refinements, and sharp total-variation constants for the inactive-vertex count, including an improved tilted Poisson reference. Explicit finite smooth models yield controlled Newton and convergent local Lagrange inverses, arbitrary fixed index accuracy, and a two-ceiling envelope for exact thresholds. The proofs include global tail bounds and never differentiate an unspecified discrete remainder. A self-contained exact-arithmetic package accompanies the article.

Scope. All asymptotic assertions keep p and the requested order fixed. No secondary exponential sector, optimal truncation theorem, growing- p limit, effective finite inverse onset, or critical marked-law theorem is asserted.

1 Results and reading guide

The family under consideration is defined for integers $p \geq 1$ and $n \geq 0$ by

$$a_p(n) = \sum_{k=0}^{\lfloor n/q \rfloor} (n - pk)^{pk} \binom{n - pk}{k}, \quad q = p + 1. \quad (1.1)$$

We set the $k = 0$ summand equal to one, including at $n = 0$. Thus $a_p(0) = 1$; no normalization involving n/q is used at zero. The cases $p = 1, 2, 3$ are the exact finite sums in A360592, A360479, and A360747 respectively [1, 2, 3].

The key parameter is

$$c_p = \exp(p^2/q)q^{-1/q}, \quad \lambda = \lambda_p(n) = c_p n^{1/q}. \quad (1.2)$$

For $p \geq 2$, let

$$M_p(n) = \frac{1}{q}(n/q)^{pn/q}e^\lambda, \quad \mathcal{R}_p(n) = \frac{a_p(n)}{M_p(n)}. \quad (1.3)$$

The first polynomial perturbation has a particularly simple universal form:

$$\mathcal{R}_p(n) = 1 - \frac{A\lambda^2 + b\lambda}{n} + O_p\left(\frac{\lambda^4}{n^2}\right), \quad A = \frac{p^2 + 1/q}{2}, \quad b = A - \frac{q}{2} = \frac{p(p^2 - 2)}{2q}. \quad (1.4)$$

For every fixed $K \geq 0$, the stronger expansion is

$$\mathcal{R}_p(n) = \sum_{j=0}^K C_{p,j}(c_p)n^{-j/q} + O_{p,K}(n^{-(K+1)/q}), \quad (1.5)$$

where $C_{p,j}$ is an explicitly generated rational polynomial and $C_{p,0} = 1$. The coefficients $C_{p,1}, \dots, C_{p,p-2}$ vanish. Every assertion is uniform over the finitely many residue classes of n modulo q .

At $p = 1$ the quantity λ^2/n is constant, so (1.4) is not an asymptotic expansion around one. The correct normalization is instead

$$c = c_1 = \sqrt{e/2}, \quad d = 3e/8 = 3c^2/4, \quad M_1(n) = \frac{1}{2}(n/2)^{n/2} \exp(c\sqrt{n} - d). \quad (1.6)$$

Then $a_1(n)/M_1(n)$ has an expansion in $n^{-1/2}$ to every fixed order. There are no odd quarter powers. Its first correction is

$$C_{1,1}(c) = \frac{c}{4} + \frac{11c^3}{8}. \quad (1.7)$$

1.1 What is corrected

The observed OEIS formulas all use the symbol $\sim$. Interpreted literally as a leading equivalent, each remains correct. Their displayed parenthesized refinements suggest more precise coefficients; it is these intended coefficients that disagree with the proved expansions here. For $p = 1$ and $p = 2$ the observed first coefficient exceeds the correct one by $1/(8c_p)$. For $p = 3$ the observed coefficient of $n^{-1/4}$ is $1/(8c_3)$, whereas the true coefficient is zero. The first nonzero $p = 3$ correction is negative

and of order $n^{-1/2}$. These statements concern inspected or indexed source snapshots, not verified latest revision histories. Section 7.4 records the source scope explicitly.

[write] **7 October 2026.** The three entries as the write read them, with their revision numbers, and the write's checks of the transcription and of the corrected coefficients against the exact sums, are in Remark 7.1.

1.2 What the extra results add

The integer $r = n - qk$ has a useful combinatorial meaning: it counts inactive vertices in a canonically labeled directed-graph model. For fixed $p \geq 2$, its distribution is close to a Poisson distribution conditioned on $r \equiv n \pmod{q}$. We identify the sharp total-variation constants and improve the approximation by an explicit Lambert- W tilt. The conditioning cannot be dropped in total variation.

The rapidly increasing counts also have controlled inverse descriptions. An explicit finite asymptotic model, followed by finitely many Newton steps, recovers the index on the exact range to any prescribed fixed algebraic accuracy. For a general real level y , we prove an envelope involving two ceilings. It can leave two adjacent possible thresholds; arbitrary tiny real error does not justify an unqualified rounding rule.

1.3 Proof architecture and classical ingredients

Section 2 obtains an exact Poisson representation by elementary algebra. Section 3 combines global domination, a local Cauchy estimate, and the root-of-unity filter. Section 4 uses a different centered argument at $p = 1$. Section 5 proves the marked laws with weighted remainders. Section 6 treats finite smooth inverses.

The underlying tools are classical: Poisson factorial and central moments, finite-difference or Faulhaber polynomial calculus, root-of-unity filters, Chernoff bounds, analytic Taylor estimates, Charlier polynomials, Lagrange inversion, and the Lambert W function [4, 5]. We give the identities and estimates needed here so that the argument does not depend on locating a theorem with this exact sequence as an example. No first-priority claim is needed for any mathematical conclusion.

2 Exact counting and Poisson reduction

2.1 A formal generating function and a graph model

Expanding the finite binomial at each m gives the formal identity

$$\sum_{n \geq 0} a_p(n) x^n = \sum_{m \geq 0} x^m (1 + (mx)^p)^m. \quad (2.1)$$

Indeed, the term indexed by k on the right is $\binom{m}{k} m^{pk} x^{m+pk}$. Setting $m = n - pk$ gives (1.1), and $k \leq m$ becomes $qk \leq n$. Every coefficient receives only finitely many contributions, so no analytic convergence is being claimed. In fact, the coefficients have superexponential growth.

For a direct interpretation, use the canonical labeled vertex set $[m] = \{1, \dots, m\}$. Choose k active vertices. Each active vertex has p ordered, or color-labeled, outgoing arc slots, and each slot

independently chooses a target in $[m]$. Loops and repeated targets are allowed. Inactive vertices have no outgoing slots. There are $\binom{m}{k} m^{pk}$ such structures. The size statistic is

$$n = m + pk = \text{number of vertices plus number of arcs.}$$

The disjoint union over the possible m therefore has size $a_p(n)$. This is a labeled model with a nonstandard size statistic; (2.1) is not an ordinary unlabeled-digraph generating function.

The number of inactive vertices is

$$r = m - k = n - qk, \quad k = \frac{n - r}{q}, \quad m = \frac{n + pr}{q}. \quad (2.2)$$

For a uniformly chosen structure of size n , denote this random number by R_n . Its support is exactly $\{r : 0 \leq r \leq n, r \equiv n \pmod{q}\}$.

2.2 An exact change of variables

For $n > 0$ write $N = n/q$ and $B = N^{pn/q}$. For an allowed r , use $\binom{m}{k} = \binom{m}{r}$ to get

$$m^{pk} \binom{m}{k} = \frac{m^{pk+r}}{r!} \prod_{h=0}^{r-1} \left(1 - \frac{h}{m}\right).$$

Since $pk + r = (pn + r)/q$ and $m = N(1 + pr/n)$, this becomes

$$m^{pk} \binom{m}{k} = B \frac{\lambda^r}{r!} e^{\Delta_n(r)}, \quad (2.3)$$

where

$$\Delta_n(r) = \frac{pn + r}{q} \log(1 + pr/n) - \frac{p^2 r}{q} + \sum_{h=0}^{r-1} \log \left(1 - \frac{qh}{n + pr}\right). \quad (2.4)$$

All factors inside the logarithms are positive for integers $0 \leq r \leq n$: $n + pr - q(r-1) = n - r + q > 0$. The empty product and empty sum at $r = 0$ are one and zero respectively. The same expression defines a real function at every integer in this range, not only those satisfying the congruence.

If $X \sim \text{Pois}(\lambda)$, summation gives the exact identity

$$qe^{-\lambda} \frac{a_p(n)}{B} = q \mathbb{E} \left[e^{\Delta_n(X)} \mathbf{1}_{\{X \leq n\}} \mathbf{1}_{\{X \equiv n \pmod{q}\}} \right]. \quad (2.5)$$

For $p \geq 2$ the left side is $\mathcal{R}_p(n)$. For $p = 1$ it is $e^{-d} a_1(n)/M_1(n)$. The factor q is the lattice spacing of the residue condition. It does not come from a Stirling approximation; no approximation has yet been used.

2.3 Poisson polynomial identities

Let T_j denote the Touchard polynomial,

$$T_j(z) = \sum_{\ell=0}^j \left\{ \begin{matrix} j \\ \ell \end{matrix} \right\} z^\ell, \quad \mathbb{E} X^j = T_j(\lambda). \quad (2.6)$$

Here the braces are Stirling numbers of the second kind. This follows by writing X^j in the falling-factorial basis and using $\mathbb{E}(X)_\ell = \lambda^\ell$. In particular, for each fixed j , $\mathbb{E}X^j = O_j((1 + \lambda)^j)$.

Put $\omega = e^{2\pi i/q}$ and $\kappa_q = 1 - \cos(2\pi/q) > 0$. For any integer residue a and any nonnegative integer j ,

$$q\mathbb{E}[X^j \mathbf{1}_{\{X \equiv a \pmod{q}\}}] = \sum_{\ell=0}^{q-1} \omega^{-a\ell} e^{\lambda(\omega^\ell - 1)} T_j(\lambda \omega^\ell). \quad (2.7)$$

To see this, insert $q\mathbf{1}_{\{X \equiv a\}} = \sum_{\ell=0}^{q-1} \omega^{\ell(X-a)}$ and use the Poisson generating function. The $\ell = 0$ term is $T_j(\lambda)$; all other terms are $O_{q,j}((1 + \lambda)^j e^{-\kappa_q \lambda})$. For a fixed polynomial, residue restriction therefore changes its Poisson average only by an exponentially small absolute error. This statement concerns polynomial averages. It does not identify a relative secondary exponential sector of the full nonpolynomial sum.

2.4 Exact monotonicity

Proposition 2.1. *For every $p \geq 1$,*

$$a_p(0) = \cdots = a_p(p) = 1, \quad a_p(p+1) = 2,$$

and $a_p(n+1) > a_p(n)$ for every $n \geq p$. Consequently, for each real $y > 1$ the threshold

$$T_p(y) = \min\{n \geq 0 : a_p(n) \geq y\} \quad (2.8)$$

is well-defined.

Proof. For each previously allowed k , increasing n increases $m = n - pk \geq k$. Both m^{pk} and $\binom{m}{k}$ are nondecreasing in m , and all newly allowed summands are positive. At the first step $n = p$, the new $k = 1$ term gives strictness. Thereafter the common $k = 1$ term $(n - p)^{p+1}$ strictly increases. It also proves unboundedness. The initial equalities are immediate from (1.1). $\square$

3 Every fixed order for the small deformation regime

Throughout this section $p \geq 2$ is a fixed integer. Thus $\lambda \asymp_p n^{1/q}$ and

$$\lambda^2/n \asymp_p n^{-(p-1)/q} \longrightarrow 0. \quad (3.1)$$

This is the small parameter that makes an ordinary exponential expansion of the defect valid after Poisson averaging.

3.1 Global domination and tails

Lemma 3.1. *For every integer $0 \leq r \leq n$, $\Delta_n(r) \leq 0$. Consequently*

$$0 < \mathcal{R}_p(n) \leq q\mathbb{P}(X \equiv n \pmod{q}) = 1 + O_p(e^{-\kappa_q \lambda}). \quad (3.2)$$

Proof. Let

$$F(r) = \frac{pn + r}{q} \log(1 + pr/n) - \frac{p^2 r}{q}.$$

Direct differentiation gives $F(0) = F'(0) = 0$ and

$$F''(r) = \frac{p(2 - p^2 + pr/n)}{qn(1 + pr/n)^2}. \quad (3.3)$$

For $0 \leq r \leq n$, its numerator is at most $p(2 - p^2 + p) \leq 0$. Hence $F(r) \leq 0$. Every logarithm in the sum in (2.4) is nonpositive. This proves the sign. Now apply (2.5) and (2.7) with $j = 0$. $\square$

The right side of (3.2) need not be identically one: residue probabilities may have exponentially small oscillations. Nevertheless it already excludes a positive algebraic first correction. In particular, a positive $n^{-1/4}$ coefficient for $p = 3$ is impossible.

The Poisson moment generating function gives, with $h = 2 \log 2 - 1 > 0$,

$$\mathbb{P}(X > 2\lambda) \leq e^{-h\lambda}. \quad (3.4)$$

Indeed, exponential Markov with parameter $\log 2$ bounds the probability by $\exp(\lambda(2 - 1) - 2\lambda \log 2)$. For each fixed integer $d \geq 0$, Cauchy–Schwarz yields

$$\mathbb{E}[X^d \mathbf{1}_{\{X > 2\lambda\}}] \leq (\mathbb{E}X^{2d})^{1/2} \mathbb{P}(X > 2\lambda)^{1/2} = O_d((1 + \lambda)^d e^{-h\lambda/2}). \quad (3.5)$$

By Lemma 3.1, the exact normalized contribution outside $X \leq 2\lambda$ is bounded by the same unweighted tail, up to q . For large n , $2\lambda < n$, so the original cutoff at n causes no additional difficulty. These are global estimates, including the far endpoint $r = n$.

3.2 Defect polynomials and a uniform analytic remainder

For $j \geq 1$ define

$$D_j(r) = (-1)^{j+1} \left\{ \frac{p^j}{q} \left(\frac{1}{j} - \frac{p^2}{j+1} \right) r^{j+1} + \frac{1}{j} \sum_{h=0}^{r-1} [(pr - qh)^j - (pr)^j] \right\}. \quad (3.6)$$

The finite sum is interpreted as its polynomial in r , obtained for example from Faulhaber sums or interpolation in the falling-factorial basis. Thus $D_j \in \mathbb{Q}[r]$, $D_j(0) = 0$, and $\deg D_j \leq j + 1$. Define

$$H_0(r) = 1, \quad H_j(r) = \frac{1}{j} \sum_{s=1}^j s D_s(r) H_{j-s}(r). \quad (3.7)$$

Formal differentiation of an exponential shows that

$$\exp \left(\sum_{j \geq 1} D_j(r) z^j \right) = \sum_{j \geq 0} H_j(r) z^j.$$

Induction gives $\deg H_j \leq 2j$ and $H_j(0) = 0$ for $j \geq 1$.

Lemma 3.2. *For each fixed $S \geq 0$, uniformly over integers $0 \leq r \leq 2\lambda$,*

$$e^{\Delta_n(r)} = \sum_{j=0}^S H_j(r) n^{-j} + O_{p,S}((r+1)^{2S+2} n^{-S-1}). \quad (3.8)$$

Proof. For an integer $r \geq 1$ consider the analytic germ

$$G_r(z) = e^{-p^2 r/q} (1 + prz)^{p/(qz) - pr/q} \prod_{h=0}^{r-1} (1 + (pr - qh)z), \quad (3.9)$$

using logarithmic branches at $z = 0$. The apparent singularity at zero is removable and $G_r(0) = 1$. Recombining the denominators in (2.4) gives $G_r(1/n) = e^{\Delta_n(r)}$.

Choose a small constant $\eta_p > 0$ and put $\rho_r = \eta_p/(r+1)^2$. For $|z| \leq \rho_r$, every logarithm argument in (3.9) lies in a fixed disk about one excluding zero. After its constant $p^2 r/q$ is subtracted, the expression $(p/(qz)) \log(1 + prz)$ is $O_p(r^2|z|)$. The remaining logarithm term and the sum of the r product logarithms are also $O_p(r^2|z|)$. Therefore

$$|\log G_r(z)| \leq C_p r^2 |z| \leq C'_p, \quad |G_r(z)| \leq e^{C'_p},$$

uniformly in r . Expanding its logarithm gives exactly (3.6); exponentiation gives (3.7).

Cauchy's coefficient estimate on $|z| = \rho_r$ and a geometric tail give

$$\left| G_r(1/n) - \sum_{j=0}^S H_j(r) n^{-j} \right| \leq C_{p,S} (r+1)^{2S+2} n^{-S-1}$$

whenever $(r+1)^2/n \leq \eta_p/2$. This condition holds uniformly for $r \leq 2\lambda$ by (3.1). For $r = 0$, $G_0 = 1$ and the conclusion is immediate. $\square$

3.3 The coefficient theorem

Define the Poisson transform of H_j by

$$H_j(r) = \sum_d h_{j,d} r^d, \quad U_j(z) = \sum_d h_{j,d} T_d(z). \quad (3.10)$$

Thus $U_0 = 1$, $U_j(0) = 0$ for $j \geq 1$, and $\deg U_j \leq 2j$.

Theorem 3.3. *For every fixed $S \geq 0$,*

$$\mathcal{R}_p(n) = \sum_{j=0}^S n^{-j} U_j(\lambda) + O_{p,S} \left(\frac{(1+\lambda)^{2S+2}}{n^{S+1}} + e^{-c_{p,S}\lambda} \right) \quad (3.11)$$

for some $c_{p,S} > 0$. In particular the exponential term can be absorbed into the algebraic remainder. Setting $t = n^{-1/q}$ and $\lambda = c_p/t$, for every fixed $K \geq 0$ this implies (1.5), with

$$C_{p,k}(c) = [t^k] \sum_{j=0}^{\lfloor K/(p-1) \rfloor} t^{qj} U_j(c/t), \quad 0 \leq k \leq K. \quad (3.12)$$

Proof. Apply (3.8) in the exact expectation on $X \leq 2\lambda$. The expectation of the remainder is at most a constant times $n^{-S-1}\mathbb{E}(X+1)^{2S+2}$, giving the algebraic term in (3.11). The exact tail is controlled by (3.4) and the sign lemma. The tails of the finitely many polynomials H_j are controlled by (3.5). Remove their residue restrictions using (2.7). All resulting exponential terms, including fixed polynomial factors, can be combined into $O(e^{-c_p, S\lambda})$.

A monomial $n^{-j}\lambda^d$ becomes $c^d t^{qj-d}$, with $1 \leq d \leq 2j$ when $j \geq 1$. Its possible exponent lies in

$$[(p-1)j, qj-1]. \quad (3.13)$$

Take $S = \lfloor K/(p-1) \rfloor$. The remainder in (3.11) starts at order $t^{(p-1)(S+1)} = O(t^{K+1})$. Discarding finitely many polynomial terms with exponent larger than K also gives $O(t^{K+1})$. This proves (3.12) and the theorem. $\square$

Formula (3.13) gives a structural exclusion of low powers. It does not imply that every exponent inside an allowed band has a nonzero coefficient. In particular, the absence of $t, t^2, \dots, t^{p-2}$ requires no symbolic computation.

3.4 Explicit corrected terms

The first two defect polynomials are

$$D_1(r) = -Ar^2 + \frac{q}{2}r, \quad (3.14)$$

$$D_2(r) = \gamma_3 r^3 - \gamma_2 r^2 - \gamma_1 r, \quad (3.15)$$

$$\gamma_3 = \frac{p^3}{3} - \frac{1}{6q}, \quad \gamma_2 = \frac{p^2 - 1}{4}, \quad \gamma_1 = \frac{q^2}{12}.$$

Since $U_1(\lambda) = \mathbb{E}D_1(X) = -A\lambda^2 - b\lambda$, Theorem 3.3 gives (1.4). For $p = 2$, write $c = e^{4/3}/3^{1/3}$ and $t = n^{-1/3}$. Then

$$\begin{aligned} \mathcal{R}_2(n) &= 1 - \frac{13c^2}{6}t + \left(\frac{169c^4}{72} - \frac{2c}{3} \right) t^2 \\ &\quad + \left(\frac{121c^3}{9} - \frac{2197c^6}{1296} \right) t^3 + O(t^4). \end{aligned} \quad (3.16)$$

One additional coefficient is

$$C_{2,4}(c) = \frac{134c^2}{9} - \frac{2977c^5}{108} + \frac{28561c^8}{31104}.$$

For $p = 3$, write $c = e^{9/4}/\sqrt{2}$ and $t = n^{-1/4}$. Then

$$\begin{aligned} \mathcal{R}_3(n) &= 1 - \frac{37c^2}{8}t^2 - \frac{21c}{8}t^3 + \frac{1369c^4}{128}t^4 + O(t^5) \\ &= 1 - \frac{37e^{9/2}}{16}n^{-1/2} - \frac{21e^{9/4}}{8\sqrt{2}}n^{-3/4} + \frac{1369e^9}{512}n^{-1} + O(n^{-5/4}). \end{aligned} \quad (3.17)$$

The next coefficients are

$$C_{3,5}(c) = \frac{12265c^3}{192}, \quad C_{3,6}(c) = \frac{9471c^2}{128} - \frac{50653c^6}{3072}.$$

All these identities are rational-polynomial outputs of the finite recurrence, not numerical coefficient fits.

Remark 3.4 (Formal order and useful numerical accuracy). The theorem is asymptotic for each fixed truncation. It does not say that larger order improves every finite- n approximation. The constants in (3.17) are large. At $n = 10000$, even the fractional-power truncation through t^{12} has relative error about -0.662 . Section 7.2 gives the complete-sum diagnostic and contrasts it with another legitimate finite truncation of (3.11).

4 The critical half power expansion

In this section $p = 1$, $q = 2$, $c = \sqrt{e/2}$, $t = n^{-1/2}$, and $\lambda = c/t$. The normalization M_1 is (1.6). Since $\lambda^2/n = c^2$, the leading quadratic defect persists: $\Delta_n(X)$ is approximately $-3(tX)^2/4$, whose limiting value is $-3c^2/4$. Expanding $e^{\Delta_n(X)}$ as a small perturbation of one would miss an order-one factor. We retain that factor and expand around the Poisson center instead.

Theorem 4.1. *For every fixed integer $K \geq 0$, there are explicit polynomials $C_{1,j} \in \mathbb{Q}[c]$ such that*

$$\frac{a_1(n)}{M_1(n)} = \sum_{j=0}^K C_{1,j}(c) n^{-j/2} + O_K(n^{-(K+1)/2}). \quad (4.1)$$

Both parities have the same coefficients. There are no odd quarter powers in the normalized expansion. The first four nonconstant coefficients are

$$C_{1,1}(c) = \frac{c}{4} + \frac{11c^3}{8}, \quad (4.2)$$

$$C_{1,2}(c) = \frac{31c^2}{32} - \frac{319c^4}{96} + \frac{121c^6}{128}, \quad (4.3)$$

$$C_{1,3}(c) = -\frac{5c}{96} - \frac{739c^3}{128} + \frac{46517c^5}{3840} - \frac{7381c^7}{1536} + \frac{1331c^9}{3072}, \quad (4.4)$$

$$C_{1,4}(c) = -\frac{227c^2}{128} + \frac{61923c^4}{2048} - \frac{771061c^6}{15360} + \frac{4140367c^8}{184320} - \frac{41261c^{10}}{12288} + \frac{14641c^{12}}{98304}. \quad (4.5)$$

4.1 A separate global bound

The exact Poisson identity is still available, now in the form

$$\frac{a_1(n)}{M_1(n)} = 2e^{3c^2/4} \mathbb{E} \left[e^{\Delta_n(X)} \mathbf{1}_{\{X \leq n\}} \mathbf{1}_{\{X \equiv n \pmod{2}\}} \right]. \quad (4.6)$$

The nonpositivity statement of Lemma 3.1 is false at $p = 1$; for example, $\Delta_n(1) > 0$. The required replacement is a uniform upper bound.

Lemma 4.2. *For every integer $0 \leq r \leq n$,*

$$\Delta_n(r) \leq -\frac{r^2}{4n} + \frac{r}{2n} = \frac{1 - (r-1)^2}{4n} \leq \frac{1}{4n}. \quad (4.7)$$

Proof. The first two terms of (2.4) now form $F(r) = ((n+r)/2)\log(1+r/n) - r/2$. One has $F(0) = F'(0) = 0$ and $F''(r) = 1/(2n(1+r/n)) \leq 1/(2n)$, so $F(r) \leq r^2/(4n)$. Since $\log(1-x) \leq -x$ for $0 \leq x < 1$,

$$\sum_{h=0}^{r-1} \log\left(1 - \frac{2h}{n+r}\right) \leq -\frac{r(r-1)}{n+r} \leq -\frac{r(r-1)}{2n}.$$

The final inequality uses $r(r-1) \geq 0$, which holds at every allowed integer. Adding the bounds proves (4.7). $\square$

Let

$$\mathcal{C}_n = \{|X - \lambda| \leq \lambda^{3/4}\}. \quad (4.8)$$

The usual exponential Markov argument for the Poisson generating function implies, for a positive absolute constant b_0 and large λ ,

$$\mathbb{P}(\mathcal{C}_n^c) \leq 2 \exp\left(-\frac{\lambda^{3/2}}{2(\lambda + \lambda^{3/4})}\right) = O(e^{-b_0\sqrt{\lambda}}). \quad (4.9)$$

For completeness, the upper-tail Chernoff rate is $\lambda((1+s)\log(1+s) - s)$ at displacement $s\lambda$; integrating $\log(1+s) \geq s/(1+s)$ bounds it below by $\lambda s^2/(2(1+s))$. The lower-tail rate is at least $\lambda s^2/2$. Substitute $s = \lambda^{-1/4}$.

Lemma 4.2 bounds the exact normalized contribution on $\mathcal{C}_n^c$ by a fixed multiple of (4.9). Every fixed polynomially weighted tail is also negligible by Cauchy–Schwarz and Poisson moment bounds. Since $\lambda \asymp \sqrt{n}$, these errors are smaller than every power of n^{-1} . For large n , the central event lies inside $0 \leq X \leq 2\lambda < n$.

4.2 Defect polynomials at criticality

Specializing (3.6) gives

$$D_j(r) = (-1)^{j+1} \left\{ \frac{r^{j+1}}{2j(j+1)} + \frac{1}{j} \sum_{h=0}^{r-1} [(r-2h)^j - r^j] \right\}. \quad (4.10)$$

The first values are

$$\begin{aligned} D_1 &= -3r^2/4 + r, & D_2 &= r^3/4 - r/3, \\ D_3 &= -7r^4/24 + r^3/3, & D_4 &= 7r^5/40 - r^3/3 + 2r/15, \\ D_5 &= -11r^6/60 + r^5/5. \end{aligned}$$

We require a local logarithmic estimate on a larger analytic scale than in Lemma 3.2. For every fixed $L \geq 1$, uniformly for integers $0 \leq r \leq 2\lambda$,

$$\Delta_n(r) = \sum_{j=1}^L D_j(r) n^{-j} + O_L\left(\frac{r^{L+2}}{n^{L+1}}\right) = \sum_{j=1}^L t^{2j} D_j(r) + O_L(t^L). \quad (4.11)$$

To verify this, rewrite the defect exactly as

$$\Delta_n(r) = \frac{n-r}{2} \log(1+r/n) - \frac{r}{2} + \sum_{h=0}^{r-1} \log(1+(r-2h)/n).$$

Expand the first logarithm through degree $L + 1$ and each logarithm in the sum through degree L . Their arguments are within $O(t)$ of one. The usual logarithmic remainders, together with the n, r prefactors and the r terms in the sum, give $O(r^{L+2}/n^{L+1})$. The statement at $r = 0$ is exact. Collecting coefficients gives (4.10).

4.3 Centered moments and a finite weighted generator

Put

$$U = t(X - \lambda) = tX - c. \quad (4.12)$$

Let $B_{d,k}^{\geq 2}$ count set partitions of d labeled elements into k blocks, with no singleton blocks. They satisfy

$$B_{0,0}^{\geq 2} = 1, \quad B_{d,k}^{\geq 2} = kB_{d-1,k}^{\geq 2} + (d-1)B_{d-2,k-1}^{\geq 2}, \quad (4.13)$$

with out-of-range indices zero. The recurrence separates the block containing the last label according as it has more than two elements or has two elements. The centered Poisson generating function is

$$\mathbb{E}e^{z(X-\lambda)} = \exp(\lambda(e^z - 1 - z)).$$

Coefficient comparison gives the exact polynomial identity

$$\mathbb{E}U^d = \sum_{k=0}^{\lfloor d/2 \rfloor} B_{d,k}^{\geq 2} c^k t^{d-k}. \quad (4.14)$$

Thus only integer powers of t occur, and their least possible exponent is $\lceil d/2 \rceil$. In particular, $\mathbb{E}|U|^{2h} = O_h(t^h)$ for each fixed h .

For the chosen expansion order K , define

$$V_K(t, u) = \frac{3c^2}{4} + \sum_{j=1}^{K+1} t^{2j} D_j((c+u)/t). \quad (4.15)$$

It is a polynomial with no negative powers of t because $\deg D_j \leq j + 1$. Its constant term vanishes and

$$V_K(0, u) = -\frac{3cu}{2} - \frac{3u^2}{4}. \quad (4.16)$$

Assign weight two to t and weight one to u . Let $P_K(t, u)$ be $e^{V_K(t,u)}$ truncated to weighted degree at most $2K + 1$. Every monomial of V_K has positive weight, so it suffices to truncate the finite sum

$$\sum_{m=0}^{2K+1} \frac{V_K(t, u)^m}{m!}. \quad (4.17)$$

Replace every u^d in P_K by the right side of (4.14), and retain powers through t^K . This finite procedure defines $C_{1,0}, \dots, C_{1,K}$. It is different from the small-deformation generator of the previous section.

4.4 Remainder and parity proof

Proof of Theorem 4.1. On $\mathcal{C}_n$, one has $|U| \leq t\lambda^{3/4} = O(t^{1/4}) \rightarrow 0$. Take $L = K + 1$ in (4.11). The exponent error is $O(t^{K+1})$, and $V_K(t, U)$ is uniformly bounded on this event. Therefore

$$e^{\Delta_n(X) + 3c^2/4} = e^{V_K(t, U)} + O_K(t^{K+1}) \quad \text{on } \mathcal{C}_n. \quad (4.18)$$

Set $s = \sqrt{t}$ solely to apply ordinary multivariate Taylor's theorem to $e^{V_K(s^2, u)}$ at $(0, 0)$. Its total-degree $2K + 1$ Taylor polynomial is $P_K(s^2, u)$. The remainder is $O_K((|s| + |u|)^{2K+2})$ in a fixed neighborhood. Averaging this bound, using (4.14) and $(a + b)^m \leq 2^{m-1}(a^m + b^m)$, gives $O_K(t^{K+1})$. The parity-restricted expectation costs at most a factor two. The extra odd weighted degree is important: a Taylor argument stopped at $2K$ would give only an absolute $O(t^{K+1/2})$ bound by this method.

The polynomial tail estimate and (4.9) allow us to replace the central expectation of P_K by its unrestricted Poisson expectation. For every fixed polynomial $Q(X)$ the exact parity filter says

$$2\mathbb{E}[Q(X)\mathbf{1}_{\{X \equiv n \pmod{2}\}}] = \mathbb{E}Q(X) + (-1)^n \mathbb{E}[(-1)^X Q(X)].$$

For $Q(X) = X^j$, the last expectation is $e^{-2\lambda} T_j(-\lambda)$. A monomial $t^i U^d = t^i (tX - c)^d$ is a finite combination of $t^{i+j} X^j$. Since $t^j T_j(-c/t)$ remains bounded, the parity contribution from P_K is $O_K(e^{-2\lambda})$. Finally (4.14) produces only integer powers of t ; those beyond K are $O_K(t^{K+1})$. Combining these statements with (4.6) proves the expansion and the claimed absence of odd quarter powers. The displayed coefficients result from the finite generator. $\square$

4.5 The first coefficient without symbolic expansion

There is a short independent way to see the first correction. Put $Z = tX = c + U$ and $f(z) = e^{-3z^2/4}$. On the bulk, the first two defect polynomials give

$$e^{\Delta_n(X)} = f(Z) \{1 + t(Z + Z^3/4) + O(t^2)\}.$$

Since $\mathbb{E}U = 0$ and $\mathbb{E}U^2 = ct$, the centered Taylor calculation gives

$$\mathbb{E}f(Z) = f(c) + \frac{ct}{2} f''(c) + O(t^2).$$

Here a fourth-order Taylor remainder and the exact third central moment justify the $O(t^2)$ error; an absolute third-moment bound alone would be insufficient. The same argument applied to $f(Z)(Z + Z^3/4)$ gives its expectation equal to its value at c plus $O(t)$. The tails have already been controlled globally. Dividing by $f(c)$, the coefficient is

$$c + \frac{c^3}{4} + \frac{c}{2} \left(-\frac{3}{2} + \frac{9c^2}{4} \right) = \frac{c}{4} + \frac{11c^3}{8}.$$

This identifies the missing cancellation without a symbolic black box. It also explains why adding the observed extra $1/(8c)$ changes the true refinement while leaving the leading equivalent untouched.

5 The inactive vertex distribution

This entire section assumes fixed $p \geq 2$. Let R_n be the inactive-vertex count from Section 2, let $a \equiv n \pmod{q}$, and let $Q_{\nu,a}$ be the distribution of $\text{Pois}(\nu)$ conditioned on the residue class a modulo q . For probability measures on this lattice we use

$$d_{\text{TV}}(P, Q) = \frac{1}{2} \sum_r |P(r) - Q(r)|.$$

The exact Poisson identity shows that the mass of R_n is proportional to

$$\frac{\lambda^r}{r!} e^{\Delta_n(r)}, \quad 0 \leq r \leq n, \quad r \equiv a \pmod{q}. \quad (5.1)$$

All constants $A, b, \gamma_1, \gamma_2, \gamma_3$ retain the meanings in (1.4) and (3.15).

5.1 The two Poisson references

Theorem 5.1. *There is a unique positive solution μ to*

$$\log(\mu/\lambda) = -\frac{2A\mu + b}{n}. \quad (5.2)$$

In terms of the principal positive branch of the Lambert function,

$$\mu = \frac{n}{2A} W\left(\frac{2A\lambda}{n} e^{-b/n}\right) = \lambda - \frac{2A\lambda^2 + b\lambda}{n} + O_p(\lambda^3/n^2). \quad (5.3)$$

The mean and variance satisfy

$$\mathbb{E}R_n = \lambda - \frac{2A\lambda^2 + b\lambda}{n} + O_p(\lambda^3/n^2), \quad (5.4)$$

$$\text{Var } R_n = \lambda - \frac{4A\lambda^2 + b\lambda}{n} + O_p(\lambda^3/n^2). \quad (5.5)$$

Moreover,

$$d_{\text{TV}}(\mathcal{L}(R_n), Q_{\lambda,a}) \sim A\sqrt{2/\pi} \frac{\lambda^{3/2}}{n}, \quad (5.6)$$

$$d_{\text{TV}}(\mathcal{L}(R_n), Q_{\mu,a}) \sim A\sqrt{2/(\pi e)} \frac{\mu}{n}. \quad (5.7)$$

The estimates and equivalents hold uniformly over the residue classes.

Uniqueness in (5.2) follows because $\log \mu + 2A\mu/n$ is strictly increasing from $-\infty$ to $+\infty$. Equation (5.3) follows from $W(z)e^{W(z)} = z$; its expansion uses $\lambda/n \rightarrow 0$. Since $\lambda^2/n \rightarrow 0$, the remainder in (5.4) and (5.5) is smaller even than the displayed λ/n terms. In particular,

$$\frac{\text{Var } R_n}{\mathbb{E}R_n} = 1 - \frac{2A\lambda}{n} + O_p(\lambda^2/n^2). \quad (5.8)$$

The reference in both total-variation statements is conditioned. An unconditioned Poisson distribution gives mass approaching $1 - 1/q$ to the complement of the support of R_n , so its total variation from R_n is at least $1 - 1/q + o(1)$. A Gaussian limit after scaling and a small total-variation distance on the original integer lattice are therefore different assertions.

5.2 A density expansion with weighted remainder

Define the monic second Poisson–Charlier polynomial by

$$\mathcal{C}_2(r; \mu) = (r - \mu)^2 - r. \quad (5.9)$$

This normalization is μ^2 times the degree-two Charlier polynomial in the generating-function convention of DLMF 18.23.5 [4]. Let

$$K_2(r; \mu) = D_2(r) + \frac{A^2}{2} \mathcal{C}_2(r; \mu)^2. \quad (5.10)$$

Write $\mathbb{E}_\mu$ for ordinary Poisson expectation and $\mathbb{E}_Q$ for expectation under $Q_{\mu,a}$.

Theorem 5.2. *On the support of $Q_{\mu,a}$ define $g_n(r) = \mathbb{P}(R_n = r)/Q_{\mu,a}(r)$, taking $g_n(r) = 0$ for $r > n$. Then*

$$\begin{aligned} g_n(r) = 1 - \frac{A}{n} \mathcal{C}_2(r; \mu) \\ + \frac{K_2(r; \mu) - \mathbb{E}_\mu K_2(X; \mu)}{n^2} + \epsilon_n(r), \end{aligned} \quad (5.11)$$

where, for $s = 0, 1, 2$,

$$\mathbb{E}_Q[|X - \mu|^s |\epsilon_n(X)|] = O_p(\mu^{4+s/2}/n^3). \quad (5.12)$$

Consequently,

$$\mathbb{E}_Q \left| g_n(X) - 1 + \frac{A}{n} \mathcal{C}_2(X; \mu) \right| = O_p(\mu^{5/2}/n^2). \quad (5.13)$$

For each fixed $L > 0$, uniformly at allowed points with $|r - \mu| \leq L\sqrt{\mu}$,

$$g_n(r) = 1 - \frac{A}{n} \mathcal{C}_2(r; \mu) + O_{p,L}(\mu^{5/2}/n^2). \quad (5.14)$$

Proof. Write $\beta_0 = \log(\mu/\lambda) = -(2A\mu + b)/n$. Changing the reference Poisson parameter subtracts $\beta_0 r$ from the exponent in (5.1), up to a normalizing constant. The identity $q/2 + b = A$ gives the exact cancellation

$$\frac{D_1(r)}{n} - \beta_0 r = \frac{A\mu^2}{n} - \frac{A}{n} \mathcal{C}_2(r; \mu). \quad (5.15)$$

The constant disappears after normalization. On $r \leq 2\mu$, logarithmic Taylor expansion gives

$$\Delta_n(r) = \frac{D_1(r)}{n} + \frac{D_2(r)}{n^2} + O_p(r^4/n^3).$$

Since $\mu/\lambda \rightarrow 1$ and $\mu^2/n \rightarrow 0$, the residual exponent is uniformly $o(1)$ on this interval:

$$V(r) = -\frac{A}{n} \mathcal{C}_2(r; \mu) + \frac{D_2(r)}{n^2} + O_p(r^4/n^3). \quad (5.16)$$

Its exponential is $1 - AC_2/n + K_2/n^2$ plus a remainder.

We give the weighted estimates explicitly. For $Y = X - \mu$, the centered Poisson cumulant generating function shows that $\mathbb{E}_\mu Y^{2h} = O_h(\mu^h)$ for $\mu \geq 1$. Cauchy–Schwarz and these even moments imply, for $s = 0, 1, 2$,

$$\begin{aligned}\mathbb{E}_\mu[|Y|^s |\mathcal{C}_2|^3] &= O_p(\mu^{3+s/2}), \\ \mathbb{E}_\mu[|Y|^s |\mathcal{C}_2 D_2|] &= O_p(\mu^{4+s/2}), \\ \mathbb{E}_\mu[|Y|^s D_2^2] &= O_p(\mu^{6+s/2}), \\ \mathbb{E}_\mu[|Y|^s X^4] &= O_p(\mu^{4+s/2}).\end{aligned}$$

For example $\mathcal{C}_2 = Y^2 - Y - \mu$ and $X = \mu + Y$ reduce each assertion to a fixed collection of central moments. Conditioning changes upper bounds only by a fixed factor because $\mathbb{P}_\mu(X \equiv a \pmod{q}) \rightarrow 1/q$. The four displayed bounds control respectively the cubic exponential term, the cross term, the square of D_2 , and the exponent remainder. The D_2^2 contribution carries n^{-4} and is absorbed into $O(\mu^{4+s/2}/n^3)$ because $\mu^2/n \rightarrow 0$.

Tails beyond 2μ are exponentially small after any fixed polynomial weight. For the exact law, this follows from the original $e^{\Delta_n(r)} \leq 1$ domination under $\text{Pois}(\lambda)$, the normalizer tending to one, and $2\mu/\lambda \rightarrow 2$. For the reference it is an ordinary Chernoff bound. Thus no global pointwise estimate on the changed-reference density is being assumed.

It remains to check normalization. The exact identities $\mathbb{E}_\mu \mathcal{C}_2 = 0$ and $\mathbb{E}_\mu \mathcal{C}_2^2 = 2\mu^2$ give $\mathbb{E}_\mu K_2 = O_p(\mu^3)$. The fixed-polynomial residue correction is exponentially small by (2.7). Dividing by the normalizer subtracts $\mathbb{E}_\mu K_2/n^2$ at second order. Its cross term with $\mathcal{C}_2/n$ has weighted size $O(\mu^{4+s/2}/n^3)$, and the normalizer-square terms have size $O(\mu^{6+s/2}/n^4)$, again absorbed by the same bound. This proves (5.11)–(5.12).

Finally, $D_2 - \mathbb{E}_\mu D_2$ has L^1 norm $O_p(\mu^{5/2})$, while $\mathcal{C}_2^2 - \mathbb{E}_\mu \mathcal{C}_2^2$ has L^1 norm $O(\mu^2)$. These follow by expanding in Y and using its central moments. They prove (5.13); its third-order remainder is smaller because $\mu^{3/2}/n \rightarrow 0$. On $|r - \mu| \leq L\sqrt{\mu}$, the corresponding pointwise centered bounds hold, and the normalized third-order remainder is $O_{p,L}(\mu^4/n^3) = o(\mu^{5/2}/n^2)$. This proves (5.14). $\square$

5.3 Moments and their second correction

The polynomial identities needed to extract the moments are

$$\begin{aligned}\mathbb{E}_\mu \mathcal{C}_2 &= 0, & \mathbb{E}_\mu[(X - \mu)\mathcal{C}_2] &= 0, \\ \mathbb{E}_\mu[((X - \mu)^2 - \mu)\mathcal{C}_2] &= 2\mu^2, & \mathbb{E}_\mu[(X - \mu)\mathcal{C}_2^2] &= 4\mu^2, \\ \mathbb{E}_\mu[((X - \mu)^2 - \mu)\mathcal{C}_2^2] &= 8\mu^3 + 8\mu^2.\end{aligned}\tag{5.17}$$

They can be verified directly by the factorial moment rule. More generally, Poisson summation by parts gives for any polynomial f

$$\text{Cov}_\mu(X, f(X)) = \mu \mathbb{E}_\mu[f(X+1) - f(X)],\tag{5.18}$$

$$\begin{aligned}\text{Cov}_\mu((X - \mu)^2, f(X)) &= \mu \mathbb{E}_\mu[f(X+1) - f(X)] \\ &\quad + \mu^2 \mathbb{E}_\mu[f(X+2) - 2f(X+1) + f(X)].\end{aligned}\tag{5.19}$$

For example, $\mathbb{E}[Xf(X)] = \mu \mathbb{E}f(X+1)$ gives the first identity; using $\mathbb{E}[(X)_2 f(X)] = \mu^2 \mathbb{E}f(X+2)$ gives the second.

Multiply (5.11) by $X - \mu$ or $(X - \mu)^2$. The second-order polynomial terms are $O_p(\mu^3/n^2)$ by (5.17)–(5.19). The weighted errors are $O_p(\mu^{9/2}/n^3)$ and $O_p(\mu^5/n^3)$, respectively, both $o(\mu^3/n^2)$. The small centered mean squared is absorbed in the variance error. Consequently

$$\mathbb{E}R_n = \mu + O_p(\mu^3/n^2), \quad \text{Var } R_n = \mu - \frac{2A\mu^2}{n} + O_p(\mu^3/n^2). \quad (5.20)$$

Substitution of (5.3) proves (5.4) and (5.5).

A useful second-order form is obtained by setting $\theta = z d/dz$ and

$$\begin{aligned} L_2(z) &= (\gamma_3 + 2A^2)z^3 + (3\gamma_3 - \gamma_2 + A^2 + 2Ab)z^2 \\ &\quad + (\gamma_3 - \gamma_2 - \gamma_1 + b^2/2)z. \end{aligned} \quad (5.21)$$

Then the same weighted calculation, retaining its polynomial terms, gives

$$\mathbb{E}R_n = \lambda - \frac{2A\lambda^2 + b\lambda}{n} + \frac{(\theta L_2)(\lambda)}{n^2} + O_p(\lambda^{9/2}/n^3), \quad (5.22)$$

$$\text{Var } R_n = \lambda - \frac{4A\lambda^2 + b\lambda}{n} + \frac{(\theta^2 L_2)(\lambda)}{n^2} + O_p(\lambda^5/n^3). \quad (5.23)$$

These are conservative weighted remainder bounds. An independent way to verify the coefficient is to write $D_1 = -A(X)_2 - bX$ and use

$$L_2 = \mathbb{E}D_2 + \frac{1}{2} \text{Var}(D_1), \quad \text{Var}((X)_2) = 4\lambda^3 + 2\lambda^2, \quad \text{Cov}((X)_2, X) = 2\lambda^2.$$

It yields exactly (5.21). In the two highlighted cases,

$$\begin{aligned} p = 2: \quad L_2(z) &= 12z^3 + \frac{44}{3}z^2 + \frac{4}{3}z, \\ p = 3: \quad L_2(z) &= \frac{4967}{96}z^3 + \frac{4515}{64}z^2 + \frac{1161}{128}z. \end{aligned}$$

5.4 Sharp total variation constants

Proof of (5.6) and (5.7). Expansion at the original reference and division by its normalizer give

$$\frac{\mathbb{P}(R_n = r)}{Q_{\lambda,a}(r)} = 1 + \frac{D_1(r) - \mathbb{E}_{Q_{\lambda,a}} D_1}{n} + \rho_n(r), \quad \mathbb{E}_{Q_{\lambda,a}} |\rho_n| = O_p(\lambda^4/n^2). \quad (5.24)$$

The same local expansion and global tail proof as in Section 3 justifies this L^1 statement, including the cutoff. Under $Q_{\lambda,a}$, write $Z_\lambda = (X - \lambda)/\sqrt{\lambda}$. Then Z_λ converges to a standard normal Z , uniformly over residues, and its fixed even moments are bounded. To see convergence, apply the residue filter to the characteristic function at $t/\sqrt{\lambda}$. The zero root gives the ordinary Poisson central limit, and every nonzero root decays exponentially for bounded t . Moment bounds follow from the fixed-polynomial filter.

Expansion of D_1 about λ now gives

$$\frac{D_1(X) - \mathbb{E}_{Q_{\lambda,a}} D_1}{\lambda^{3/2}} \longrightarrow -2AZ$$

in distribution and in absolute first moment, by uniform integrability. Since $\lambda^{5/2}/n \rightarrow 0$ for $p \geq 2$, the error in (5.24) is negligible relative to $\lambda^{3/2}/n$. Thus

$$d_{\text{TV}}(\mathcal{L}(R_n), Q_{\lambda,a}) \sim \frac{2A\lambda^{3/2}}{2n} \mathbb{E}|Z| = A\sqrt{2/\pi} \frac{\lambda^{3/2}}{n}.$$

For the tilted reference, (5.13) gives

$$d_{\text{TV}}(\mathcal{L}(R_n), Q_{\mu,a}) = \frac{A}{2n} \mathbb{E}_{Q_{\mu,a}} |\mathcal{C}_2(X; \mu)| + O_p(\mu^{5/2}/n^2).$$

The same conditional central limit and uniform integrability imply $\mathcal{C}_2(X; \mu)/\mu \rightarrow Z^2 - 1$ in distribution and absolute first moment. If ϕ is the standard normal density, integration by parts on $[0, 1]$ gives $\mathbb{E}|Z^2 - 1| = 4\phi(1)$. Since $\mu^{3/2}/n \rightarrow 0$, the error is negligible, and the leading constant is $2A\phi(1) = A\sqrt{2/(\pi e)}$. $\square$

5.5 All fixed marked orders and graph consequences

The finite polynomials U_j in (3.10) give an algebraic logarithm by the recurrence

$$L_j(z) = U_j(z) - \frac{1}{j} \sum_{k=1}^{j-1} k L_k(z) U_{j-k}(z). \quad (5.25)$$

Equivalently, as a formal series in w ,

$$\log \left(1 + \sum_{j \geq 1} U_j(z) w^j \right) = \sum_{j \geq 1} L_j(z) w^j.$$

The formal coefficient of the h th cumulant of R_n is then

$$\lambda + \sum_{j \geq 1} \frac{(\theta^h L_j)(\lambda)}{n^j}. \quad (5.26)$$

This is an effective all-fixed-order generator, with the following precise remainder interpretation. Multiply the finite density expansion through J by each required fixed power r^d , apply the polynomial filter, and form the finite quotients and cumulant combinations. The unnormalized weighted error is bounded by

$$O_{p,J,d}(\lambda^{2J+2+d}/n^{J+1}).$$

For a desired fixed algebraic order, take J sufficiently large and then collect powers after setting $\lambda = c_p n^{1/q}$. This procedure justifies (5.26) to that order without assuming any general degree cancellation in L_j , and without differentiating an uncontrolled asymptotic remainder.

For clarity, an exact marked generating function can be written before any approximation:

$$\mathbb{E} e^{sR_n} = \frac{\sum_{r \equiv a \pmod{q}, r \leq n} (\lambda e^s)^r e^{\Delta_n(r)} / r!}{\sum_{r \equiv a \pmod{q}, r \leq n} \lambda^r e^{\Delta_n(r)} / r!}.$$

Its finite derivatives explain the operator θ in the coefficient rule. The error control is supplied by the weighted argument above.

Finally, if K_n and V_n are the numbers of active and total vertices, (2.2) gives the exact relations

$$\mathbb{E}K_n = \frac{n - \mathbb{E}R_n}{q}, \quad \text{Var } K_n = \frac{\text{Var } R_n}{q^2}, \quad \mathbb{E}V_n = \frac{n + p\mathbb{E}R_n}{q}, \quad \text{Var } V_n = \frac{p^2 \text{Var } R_n}{q^2}.$$

The two vertex counts are affinely determined by the same defect. Their centered Gaussian limits follow from the conditional Poisson limit and either total-variation approximation. None of the statements in this section is asserted for $p = 1$.

6 Finite smooth models and controlled inverses

This section covers every fixed integer $p \geq 1$, using the counting theorem appropriate to that p . Its subject is inversion of explicit finite smooth models. We do not differentiate the discrete sequence, nor an unspecified error in its asymptotic expansion.

6.1 Unified notation and model derivatives

Put

$$\beta = 1/q, \quad \kappa = p/q, \quad \alpha = c_p, \quad d_p = \begin{cases} 3e/8, & p = 1, \\ 0, & p \geq 2, \end{cases} \quad (6.1)$$

and use the smooth positive function

$$M_p(x) = q^{-1}(x/q)^{\kappa x} \exp(\alpha x^\beta - d_p), \quad x > 0. \quad (6.2)$$

For a fixed $K \geq 0$, define the finite sum

$$Q_K(x) = \sum_{j=0}^K C_{p,j}(\alpha) x^{-j/q}, \quad \nu_K = (K+1)/q. \quad (6.3)$$

For $p = 1$, these are the separate critical coefficients from Theorem 4.1. Both counting theorems give

$$a_p(n)/M_p(n) = Q_K(n) + O_{p,K}(n^{-\nu_K}). \quad (6.4)$$

Let

$$F_K(x) = \kappa x \log(x/q) + \alpha x^\beta - \log q - d_p + \log Q_K(x) \quad (6.5)$$

where x is sufficiently large that $Q_K(x) > 0$.

Lemma 6.1. *For each fixed p, K , there exists $X_{p,K} > e$ such that for $x \geq X_{p,K}$,*

$$Q_K(x) > 1/2, \quad F'_K(x) \geq c'_p \log x > 0, \quad 0 < F''_K(x) \leq C_{p,K}/x. \quad (6.6)$$

The function F_K has a unique smooth inverse u_K on its eventual range, and at integers

$$\log a_p(n) = F_K(n) + O_{p,K}(n^{-\nu_K}). \quad (6.7)$$

For fixed $L \geq K$ and each fixed derivative order $h \geq 0$,

$$(F_L - F_K)^{(h)}(x) = O_{p,K,L,h}(x^{-\nu_K-h}). \quad (6.8)$$

Proof. Let $\delta = 1/2$ for $p = 1$ and $\delta = (p-1)/q$ for $p \geq 2$. The vanishing coefficients already proved imply $(Q_K - 1)^{(h)}(x) = O_{p,K,h}(x^{-\delta-h})$; if $Q_K = 1$ the left side is zero. This is ordinary differentiation of a known finite expression. In particular $Q_K > 1/2$ eventually, and $(\log Q_K)^{(h)} = O(x^{-\delta-h})$. Thus

$$F'_K(x) = \kappa(\log(x/q) + 1) + \alpha\beta x^{\beta-1} + O(x^{-\delta-1}), \quad (6.9)$$

$$F''_K(x) = \kappa/x + \alpha\beta(\beta-1)x^{\beta-2} + O(x^{-\delta-2}). \quad (6.10)$$

The positive κ/x term dominates the second derivative, proving (6.6). Higher derivatives of order $h \geq 2$ are $O_{p,K,h}(x^{1-h})$. Taking a logarithm in (6.4), using $Q_K > 1/2$, proves (6.7). Finally $Q_L - Q_K$ is a finite sum starting at $x^{-\nu_K}$, so its fixed derivatives and those of the logarithmic difference have the stated bounds. This proves (6.8). $\square$

6.2 A Lambert core and finitely many Newton steps

Write $Y = \log y$ and set

$$Z = Y + \log q + d_p, \quad N = N(Y) = \frac{qZ}{pW(Z/p)}, \quad (6.11)$$

where W is the positive real branch. This solves exactly

$$\kappa N \log(N/q) = Z. \quad (6.12)$$

Indeed, $N/q = \exp(W(Z/p))$ and $W(Z/p)e^{W(Z/p)} = Z/p$. For fixed K, J , begin at $x_0 = N(Y)$ and make J Newton steps

$$x_{j+1} = x_j - \frac{F_K(x_j) - Y}{F'_K(x_j)}, \quad 0 \leq j < J. \quad (6.13)$$

This is an explicit finite composition of W , real powers, logarithms, exponentials, and arithmetic. There is no hidden root solve in x_J .

Theorem 6.2. *For every fixed p, K, J , all iterates in (6.13) are well-defined for sufficiently large Y , and*

$$u_K(Y) \leq x_{j+1} \leq x_j \leq N(Y). \quad (6.14)$$

Their error obeys

$$0 \leq x_J - u_K(Y) \leq C_{p,K,J} \frac{N^{1-(1-\beta)2^J}}{(\log N)^{2^{J+1}-1}}. \quad (6.15)$$

Proof. At the starting point, $F_K(N) - Y = \alpha N^\beta + \log Q_K(N) > 0$ eventually. On the other hand $F_K(N/2) < Y$ for large Y , because the change in the core term is of order $N \log N$, much larger than N^β . Thus $u = u_K(Y) \in (N/2, N)$. By the mean-value theorem and (6.6),

$$e_0 = N - u = O_{p,K}(N^\beta / \log N). \quad (6.16)$$

On $[u, N]$, F_K is increasing and convex. The tangent line at $x \geq u$ meets the line of height Y between u and x ; this proves the monotonicity and all domain assertions in (6.14). Taylor's theorem, expanded at x_j , gives

$$e_{j+1} = \frac{F''_K(\xi_j)}{2F'_K(x_j)} e_j^2 \leq C \frac{e_j^2}{N \log N}.$$

The power of N therefore changes by $a \mapsto 2a - 1$ and the power in the denominator by $b \mapsto 2b + 1$. Starting from $(a, b) = (\beta, 1)$ in (6.16) yields (6.15). $\square$

The first Newton error is $O(N^{2\beta-1}/\log^3 N)$. At criticality $\beta = 1/2$, so the first two Newton errors are respectively

$$O(\log^{-3} N), \quad O(N^{-1} \log^{-7} N). \quad (6.17)$$

The constant offset in the critical core is essential: $N = 2(Y + \log 2 + 3e/8)/W(Y + \log 2 + 3e/8)$.

6.3 Arbitrary fixed index accuracy

Corollary 6.3. *Given any fixed $M > 0$, choose integers $K, J \geq 0$ with*

$$(K+1)/q \geq M, \quad (1-\beta)2^J \geq M+1. \quad (6.18)$$

Let $h_{K,J}(Y) = x_J(Y)$. Then on the exact range of the count sequence,

$$h_{K,J}(\log a_p(n)) = n + O_{p,K,J}(n^{-M}/\log n). \quad (6.19)$$

In particular, its nearest integer is n for all sufficiently large n .

Proof. By (6.7) and the derivative lower bound,

$$u_K(\log a_p(n)) - n = O_{p,K}(n^{-\nu_K}/\log n).$$

The leading growth also gives $N(\log a_p(n)) \sim n$. Combine this comparison with (6.15) and (6.18). The logarithmic denominator in the Newton bound is at least $\log N$, giving (6.19). Eventually its absolute error is less than $1/2$. $\square$

The inverse models are coherent. From (6.8) and the mean-value theorem, for each fixed $L \geq K$,

$$u_L(Y) - u_K(Y) = O_{p,K,L}(N(Y)^{-\nu_K}/\log N(Y)). \quad (6.20)$$

Increasing the model order therefore cannot change already resolved inverse terms. All derivative bounds used here belong to explicit finite functions; none is inferred by differentiating (6.7).

6.4 A convergent local Lagrange generator

For a fixed sufficiently large $N = N(Y)$, put

$$v = F_K(N) - Y, \quad H_N(z) = \frac{z}{F_K(N+z) - F_K(N)}, \quad H_N(0) = \frac{1}{F'_K(N)}. \quad (6.21)$$

The local Lagrange formula is

$$u_K(Y) = N + \sum_{m \geq 1} \frac{(-v)^m}{m!} \frac{d^{m-1}}{dz^{m-1}} H_N(z)^m \Big|_{z=0}. \quad (6.22)$$

We verify a convergence disk and a quantitative truncation bound, rather than using this as a merely formal identity.

Proposition 6.4. *For fixed p, K and all sufficiently large Y , series (6.22) converges. After retaining $m = 1, \dots, L$ its remainder is*

$$O_{p,K,L} \left(N \left(\frac{N^{\beta-1}}{\log N} \right)^{L+1} \right). \quad (6.23)$$

Proof. Fix $0 < \eta < 1/4$. The explicit finite expression $F_K(N+z)$ is holomorphic on a neighborhood of $|z| \leq \eta N$ for large N : $N+z$ stays in the right half-plane, and $Q_K(N+z)$ stays near one. Uniformly on this disk,

$$F'_K(N+z) = \kappa \log N + O_{p,K,\eta}(1).$$

Its real part is positive eventually. For two points in the disk, divide their function-value difference by their point difference and integrate the derivative along their connecting segment. The real part of the result is positive, so the map is injective.

Write $G(z) = F_K(N+z) - F_K(N)$. Integrating the same derivative bound gives

$$|G(z) - \kappa(\log N)z| \leq C|z|.$$

Let $R_N = (\kappa\eta/4)N \log N$. For large N , on $|z| = \eta N$ the perturbation is at most $\frac{1}{4}\kappa\eta N \log N$. For every $|w| < 2R_N$,

$$|G(z) - \kappa(\log N)z - w| < \frac{3}{4}\kappa\eta N \log N < |\kappa(\log N)z|.$$

Rouché's theorem gives exactly one solution of $G(z) = w$ inside this disk. The derivative is nonzero there, so the inverse displacement is analytic on the open image disk $|w| < 2R_N$ and bounded by ηN . This strict radius margin permits Cauchy's estimate on $|w| = R_N$.

The Taylor coefficients of the inverse displacement are, by the local Lagrange theorem [4], $\frac{1}{m!}(d^{m-1}/dz^{m-1})H_N(z)^m|_{z=0}$. Since $v = O(N^\beta)$, one has $|v|/R_N \rightarrow 0$. Cauchy's bound and a geometric tail yield

$$O \left(N \frac{(|v|/R_N)^{L+1}}{1 - |v|/R_N} \right),$$

which proves (6.23) and convergence at $w = -v$. □

The first two displacements in (6.22) are

$$-\frac{v}{F'_K(N)} - \frac{F''_K(N)v^2}{2F'_K(N)^3}. \quad (6.24)$$

The first leading displacement alone is

$$-\frac{\alpha N^\beta}{\kappa(\log(N/q) + 1)}. \quad (6.25)$$

Discarding terms to get (6.25) also discards their accuracy. This shorter expression is not assigned the much smaller error of a multistep Newton approximation.

6.5 Exact integer thresholds and the rounding boundary

Theorem 6.5. Fix $M > 0$ and K, J satisfying (6.18). For $h = h_{K,J}(\log y)$ there are constants C, Y_0 , depending only on p, K, J , such that for $y \geq e^{Y_0}$,

$$\left\lceil h - C \frac{h^{-M}}{\log h} \right\rceil \leq T_p(y) \leq \left\lceil h + C \frac{h^{-M}}{\log h} \right\rceil. \quad (6.26)$$

The endpoint integers differ by at most one for all sufficiently large y . If they coincide, that integer is the exact threshold. Otherwise the theorem leaves two adjacent possibilities.

Proof. First use the exact model inverse $u = u_K(Y)$, $Y = \log y$. For integers m with $|m - u| \leq 2$, the error in (6.7) is bounded by $C_1 u^{-\nu_K}$ and $F'_K(x) \geq c_1 \log u$ throughout this small neighborhood. Choose

$$\varepsilon = \frac{2C_1 u^{-\nu_K}}{c_1 \log u}.$$

For $m_+ = \lceil u + \varepsilon \rceil$, $F_K(m_+) - Y \geq 2C_1 u^{-\nu_K}$, hence $\log a_p(m_+) \geq Y$. For $m_- = \lceil u - \varepsilon \rceil - 1$ one has strictly $m_- < u - \varepsilon$, even if $u - \varepsilon$ is itself an integer. It follows that $\log a_p(m_-) < Y$. Exact monotonicity from Proposition 2.1 yields

$$\lceil u - \varepsilon \rceil \leq T_p(y) \leq \lceil u + \varepsilon \rceil.$$

Replacing u by h uses (6.15); $u \sim h \sim N$ and $\nu_K \geq M$ absorb both errors into the radius in (6.26). Twice that radius is eventually less than one, so its endpoint ceilings can differ by at most one. $\square$

Remark 6.6 (Exact range versus arbitrary levels). Nearest-integer recovery in Corollary 6.3 assumes the input is exactly $y = a_p(n)$. It does not give nearest-integer threshold recovery for an arbitrary y . Even an extremely short real interval can straddle an integer, in which case taking a ceiling cannot be settled by its width alone. If actual numerical bounds and candidates are available, direct integer evaluation can decide the remaining threshold comparison. The present asymptotic theorem does not supply certified finite constants or a certified onset for doing so.

All onset thresholds and error constants in this inverse section are existential. They could in principle be made effective from explicit Taylor and concentration bounds, but the accompanying numerical inverse experiments are diagnostics, not a certified finite-input algorithm. The critical count theorem transfers to these inverses because its finite models have the same derivative structure. It does not transfer the marked probability theorem to $p = 1$.

Remark 6.7 ([write] The inverses and the transseries volume, 7 October 2026). Compared statement by statement with the repository's transseries volume [16]; the volume's symbols are not used here.

1. *The growth.* $\log a_p(n) = \kappa n \log(n/q) + \alpha n^\beta + O(1)$ contains the term $n \log n$, so it is not of the exponential–power form of `p0:def:model` (whose exponent is at most one, without a logarithmic factor); no instance of the volume's model is claimed.
2. *The Lambert core.* With the unknown replaced by its logarithm, $w = \log(N/q)$, the core equation (6.12) becomes, after one more logarithm, $w + \log w = \log(Z/p)$: the volume's dominant block `p0:thm:lambert-core` with unit slope, unit logarithmic coefficient and target $\log(Z/p)$. Its positive-coefficient branch gives exactly $w = W(Z/p)$, that is

$N = qZ/(pW(Z/p))$ of (6.11). So the core is an exact instance (an equation in N for a given level). The tilt parameter (5.3) is likewise an exact instance at fixed n : $\log \mu + (2A/n)\mu = \log \lambda - b/n$ has positive logarithmic coefficient, and its positive-branch solution is the displayed Lambert formula.

3. *The Newton iterates and the Lagrange series.* In the chart $\zeta = \log x$, with the logarithm of $Y + \log q + d_p$ as target, the leading balance of $F_K(x) = Y$ is a monomial–logarithmic datum (`plt:def:lw-monomial-log-datum`) with exponent one, logarithmic power one and constant $\log \kappa$, and the correction $\log(1 - (\log q)/\zeta)$ is a power series in $1/\zeta$. But F_K also contains αx^β , which in that chart contributes a multiple of $e^{-(1-\beta)\zeta}/\zeta$: a term beyond every power of $1/\zeta$, so the hypothesis that the perturbation be a series in $1/\zeta$ with coefficients polynomial in $\log \zeta$ fails. The write therefore has not shown the iterates (6.13) or the local series (6.22) to be instances of `plt:thm:lw-template`; they are proved directly, with explicit Newton and Cauchy bounds.
4. *The thresholds.* $T_p(y)$ is the volume’s staircase (`p0:def:three-inverses(1)`) of the eventually strictly increasing sequence $a_p(n)$ (Proposition 2.1). The smooth model F_K is not an interpolation of $\log a_p(n)$ (it differs from it by the vertical error of (6.7)), so Theorem 6.5 is proved directly: its two-ceiling envelope is an analogue of `p0:thm:staircase(2)`, not an instance, and Corollary 6.3 is an analogue of `p0:thm:staircase(3)`; the remark after Theorem 6.5 is the caution of that theorem.

7 Computation and source comparison

7.1 Exact arithmetic and independent checks

The accompanying package separates three kinds of evidence:

1. Integer and rational identities are checked with Python integers and `Fraction`. These include exact counts, defect polynomials, Poisson transforms, marked coefficients, and critical coefficients.
2. Transcendental evaluations use optional `mpmath` arithmetic. They corroborate the theorems and expose finite- n limitations; they are not certified interval enclosures.
3. The proofs above provide asymptotic remainders and quantifiers. Neither fitted coefficients nor numerical agreement replaces those proofs.

The exact core requires no symbolic algebra package. It computes finite power sums by telescoping, the H_j recurrence by rational polynomial arithmetic, and Poisson transforms by the Touchard recurrence. The critical generator separately performs weighted polynomial truncation and exact centered-moment substitution.

Independent checks reconstruct polynomial expectations in the falling-factorial basis. If a polynomial H has degree at most d , then

$$H(r) = \sum_{j=0}^d \Delta^j H(0) \binom{r}{j}, \quad \mathbb{E}H(X) = \sum_{j=0}^d \Delta^j H(0) \frac{\lambda^j}{j!}, \quad (7.1)$$

where $\Delta H(r) = H(r+1) - H(r)$. This is a distinct route from converting ordinary powers through Touchard polynomials. Extra integer samples check the interpolated identities. The public tests also multiply the formal generating function independently at small sizes, check every residue for $p = 1, 2, 3, 4$ in numerical identity tests, and test active input guards under optimized Python.

Program limits are deliberately finite resource safeguards, not restrictions of the theorems. For example, the public generator caps the formal H order at 16, fractional order at 32 subject to that cap, and critical order at 8. All guards use ordinary conditionals and exceptions, so Python's `-O` flag cannot remove them. The precise limits and commands are listed in the code README.

7.2 Complete sums and a visible preasymptotic warning

For the complete-sum diagnostic, the original expression (1.1) is evaluated as an exact integer. Its normalization uses 90-digit arithmetic. Independently, the transformed expression sums *every* allowed r from zero through n , using logarithms and log-gamma evaluations. No Poisson window and no asymptotic coefficients enter either finite-sum comparison.

For $p \geq 2$ let

$$A_K(n) = \sum_{k=0}^K C_{p,k}(c_p) n^{-k/q}, \quad S_J(n) = \sum_{j=0}^J U_j(\lambda) n^{-j}.$$

These are different finite approximations. In particular, S_{12} retains many fractional powers beyond those in A_{12} . Table 1 reports relative errors as approximation divided by $\mathcal{R}_p(n)$ minus one. The number of printed digits is deliberately smaller than the working precision.

p	n	$\mathcal{R}_p(n)$	$A_{12}/\mathcal{R}_p - 1$	$S_{12}/\mathcal{R}_p - 1$
2	10000	0.5075173547743	$-9.64017 \cdot 10^{-9}$	$+6.40647 \cdot 10^{-10}$
3	10000	0.1416076329487	-0.6615364472	$+0.0012291163$

Table 1: Complete-sum diagnostics. Both parameterizations are asymptotically valid at fixed order, but have sharply different errors here.

The original and transformed evaluations agree to about 86 decimal places in these cases. Such agreement checks the exact transformation numerically; it does not certify the final transcendental digits by outward rounding. Most importantly, the $p = 3$ fractional truncation is still wrong by about 66% at $n = 10000$. The shorter t^2 and t^4 approximations are worse. Large constants such as $e^{9/2}$ delay the useful asymptotic regime. The smaller error of S_{12} at this input does not establish a general optimal truncation principle.

For the critical case, the complete count at $n = 10000$ gives

$$a_1(n)/M_1(n) \approx 1.02446126439771172897.$$

The half-power truncation through $C_{1,4}$ has relative error approximately $-2.32029 \cdot 10^{-9}$. The companion odd input $n = 10001$ gives an error of the same size, consistent with the common algebraic coefficients for both parities.

[write] The decimals recomputed, 7 October 2026. The write recomputed Table 1 and the critical values from the exact integers at 110 digits, with its own coefficients $C_{p,k}$, $k \leq 12$, and transforms U_j , $j \leq 12$: every

printed value is the correct rounding of the recomputed one. Several are roundings rather than truncations; the truncations at the printed length are $\mathcal{R}_2 = 0.5075173547742\dots$, $A_{12}/\mathcal{R}_2 - 1 = -9.64016\dots \cdot 10^{-9}$, $S_{12}/\mathcal{R}_2 - 1 = +6.40646\dots \cdot 10^{-10}$, $\mathcal{R}_3 = 0.1416076329486\dots$, $A_{12}/\mathcal{R}_3 - 1 = -0.6615364471\dots$, $S_{12}/\mathcal{R}_3 - 1 = +0.0012291162\dots$, $a_1(n)/M_1(n) = 1.02446126439771172896\dots$ and the relative error $-2.32028\dots \cdot 10^{-9}$ at $n = 10000$ ($-2.31971\dots \cdot 10^{-9}$ at $n = 10001$). The integer and transformed evaluations agree to 10^{-106} in the write's computation.

7.3 Marked and inverse diagnostics

At $n = 10^9$ the reported TV distances divided by their proved leading equivalents are approximately

p	Original conditioned reference	Tilted conditioned reference
2	1.000111734	1.000214479
3	1.000578191	0.999189636

These computations use central windows, with the window endpoints and number of lattice points included in the machine-readable output. Chernoff bounds control the difference between the window-normalized TV and complete-law TV. Their high-precision evaluated bounds are tiny, but are still not interval certificates. The moment outputs are explicitly window diagnostics; a TV bound alone does not control unbounded moment errors.

The Newton diagnostic has two modes. By default it sets $Y = F_K(n)$, so the finite model root is known exactly to be n ; this isolates iteration error. An optional exact-target mode instead uses $Y = \log a_p(n)$ and therefore includes model truncation error. The program makes exactly the requested finite number of Newton steps and checks numerical domain conditions at each step. Neither mode supplies a certified convexity onset or certifies threshold rounding. These distinctions matter when interpreting a tiny printed residual.

[write] **The marked ratios recomputed, 7 October 2026.** With the exact law over a full central window (log-gamma values of the summands) and the residue-conditioned Poisson references, the write obtains the four ratios at $n = 10^9$ as $1.00011173445\dots$, $1.00021447898\dots$ ($p = 2$) and $1.00057819116\dots$, $0.999189635789\dots$ ($p = 3$): the printed values are correct roundings (truncations 1.000214478 and 0.999189635 for the two that differ).

7.4 Observed primary formulas

The exact finite sums and the three displayed OEIS expressions were observed in primary-entry text on 5 October 2026. The A360479 ordinary page was retrieved directly; A360592 and A360747 were obtained through indexed primary OEIS text when direct opening was unreliable. Independent retrievals corroborated the mathematical formulas. No verified sequence revision identifier or complete history export was obtained. Global page footers and indexed modification dates do not establish the current sequence revision.

A360592 identifies Václav Kotešovec as the sequence author, dated 13 February 2023. A360479 and A360747 identify Seiichi Manyama as sequence author, dated 19 February 2023; their refinement formulas are attributed to Kotešovec on 19 and 20 February 2023 respectively. These are observed source attributions, not a reconstructed publication history. The inspected entry links were numerical tables or graphs rather than proof articles.

Using the normalizations of this article, the observed critical multiplier was

$$1 + \left(\frac{c}{4} + \frac{11c^3}{8} + \frac{1}{8c} \right) t, \quad c = \sqrt{e/2}, \quad t = n^{-1/2}.$$

For $p = 2$, it was

$$\begin{aligned} & 1 + \left(-\frac{13c^2}{6} + \frac{1}{8c} \right) t \\ & + \left(\frac{169c^4}{72} - \frac{15c}{16} + \frac{67}{384c^2} \right) t^2 \\ & + \left(-\frac{2197c^6}{1296} + \frac{7913c^3}{576} + \frac{3929}{2304} + \frac{497}{3072c^3} \right) t^3, \end{aligned}$$

with $c = e^{4/3}/3^{1/3}$ and $t = n^{-1/3}$. For $p = 3$, it was

$$\begin{aligned} & 1 + \frac{t}{8c} + \left(-\frac{37c^2}{8} + \frac{67}{384c^2} \right) t^2 \\ & + \left(-\frac{205c}{64} + \frac{497}{3072c^3} \right) t^3 \\ & + \left(\frac{1369c^4}{128} + \frac{10721}{3072} + \frac{218831}{1474560c^4} \right) t^4, \end{aligned}$$

with $c = e^{9/4}/\sqrt{2}$ and $t = n^{-1/4}$. Each appeared as part of a formula using $\sim$. Every one of these multipliers tends to one, so the formulas remain correct if read solely as leading equivalents. Their intended refined coefficients differ from (4.2), (3.16), and (3.17). Appendix A lists the differences explicitly. No explanation of how the discrepancies arose is assumed.

[write] **7 October 2026.** The write compared these three multipliers with the formulas of the current entries (revisions in Remark 7.1): after the normalizations of this article they are exactly the displayed expressions (checked symbolically, leading factors included).

7.5 Bounded historical scope

The source comparison establishes a specific coefficient audit, not a universal priority result. Exact-number and formula-oriented searches did not locate a proof of the observed refinements. Two inspected default-branch catalogues in the public ProveIt repository and exact code searches did not identify this endpoint family. Those are bounded negative checks; they cannot rule out an unnamed general theorem or material elsewhere.

A potentially relevant 2013 note by Kotešovec, *Interesting asymptotic formulas for binomial sums*, was bibliographically identified at https://www.kotesovec.cz/math_articles/kotesovec_interesting_asymptotic_formulas.pdf, but retrieval timed out and its theorem content was not compared. Likewise the latest OEIS histories were not verified. Consequently no assertion that all prior relevant work has been excluded is made here. The correction follows from the exact finite sums and the self-contained proofs, regardless of broader historical priority.

For classical background, the DLMF inverse-function and Charlier generating function entries were checked directly. The Lambert- W reference was verified bibliographically and at its definition-level abstract; we do not attribute an uninspected theorem in that paper to the present family. The analytic inverse estimates needed here are proved in Section 6.

Remark 7.1 ([write] The OEIS entries, 7 October 2026). The write read the three entries in the internal format on 7 October 2026. *A360592*, revision #28 (17 February 2023), by Václav Kotešovec (13 February 2023), “G.f.: $\text{Sum}_{\{k \geq 0\}} (1 + k^*x)^k * x^k$.”, with the finite sum, a b-file by Kotešovec for $n = 0, \dots, 760$, and the unsigned formula line $a(n) \sim \exp(\exp(1/2)\sqrt{n/2} - 3e/8) n^{n/2}/2^{n/2+1} (1 + ((e^{1/2} + e^{-1/2})/2^{5/2} + 11e^{3/2}/2^{9/2})/\sqrt{n})$. *A360479*, revision #40 (19 February 2023), by Seiichi Manyama (19 February 2023), with a b-file by Kotešovec ($n = 0, \dots, 635$), a graph of the asymptotic ratio, and the refinement through n^{-1} signed by Kotešovec, 19 February 2023. *A360747*, revision #22 (20 February 2023), by Manyama (19 February 2023), with a b-file by Kotešovec ($n = 0, \dots, 600$), a graph, and the refinement through n^{-1} signed by Kotešovec, 20 February 2023, ending “see graph for more minor asymptotic terms”. These are the latest revisions, all from February 2023, so the formulas the source compared in Section 7.4 are those of the current entries, and its attributions are as stated there. *Checks*. All terms of the three b-files equal the write’s evaluation of (1.1). Rewritten in this article’s normalizations, the three formula lines are exactly the multipliers displayed in Section 7.4, with leading factors $M_1(n)$, $M_2(n)$, $M_3(n)$ (SymPy). The leading equivalents are therefore correct. The first corrections are wrong: by Theorems 4.1 and 3.3 the true first coefficients are $C_{1,1}$, $C_{2,1} = -13c^2/6$ and $C_{3,1} = 0$, and each entry adds $1/(8c_p)$. Against the exact sums the write finds (entry multiplier $-a_p(n)/M_p(n)/t$ equal to $0.10722073\dots$ at $n = 10^{14}$ ($p = 1$; $1/(8c) = 0.10722048\dots$), $0.04752092\dots$ at $n = 10^{18}$ ($p = 2$; $1/(8c_2) = 0.04752160\dots$) and $0.01863213\dots$ at $n = 10^{24}$ ($p = 3$; $1/(8c_3) = 0.01863212\dots$), the full refined multipliers being used; so the entries’ first corrections are refuted by the proofs of this Part and, numerically, by the exact sums. No OEIS edit is part of this work.

8 Further questions and boundaries of the results

The finite-order method leaves several natural questions. Each requires additional arguments beyond the theorems proved here.

Effective inverse constants. The local analytic bounds and Chernoff inequalities are explicit in form, but this article has not optimized their constants or implemented a certified finite onset. A useful next step is an interval-arithmetic inverse with a provable error radius, followed by exact count comparisons for any ambiguous adjacent thresholds.

A better moderate size approximation. The $p = 3$ diagnostic suggests retaining more of the defect in exponential form rather than immediately collecting fractional powers. The full H truncation is already much better at one displayed input, but this alone proves no generally superior scheme. Uniform relative error bounds for an appropriately resummed approximation would be more informative than choosing a truncation by appearance.

Critical marked laws. At $p = 1$ the leading quadratic tilt is order one in the partition function. The count proof resums it, but the sharp marked-law argument in Section 5 relies on $\lambda^2/n \rightarrow 0$. A separate centered analysis is needed to determine the best conditional reference and sharp distances at criticality. Nothing in the inverse transfer proves those probability results.

[write] **Answered in Part II, 7 October 2026.** Part II (Report 235) carries out this centered analysis for the inactive count: the sharp distances to both conditioned Poisson references (Corollary 11.4, with the constants of (5.6) and (5.7)), all fixed-order marked cumulants (Theorem 11.2), and the globally optimal parity-conditioned binomial with its sharp constant (Theorem 9.1). It does not determine a best reference outside the conditioned binomial family.

Residue effects beyond algebraic order. The polynomial root filter displays exponentially small terms at nonzero roots of unity. A remainder that is merely algebraically small can swamp those terms. Establishing a secondary sector of the full deformed sum would require a direct complex or oscillatory remainder analysis with relative accuracy on that smaller scale. No such claim follows from (2.7) alone.

Large order and growing parameters. All estimates hold at every chosen fixed order, with constants depending on that order and on p . They give neither convergence of the infinite asymptotic series nor optimal truncation. The scale c_p grows rapidly with p , so estimates uniform for a growing p would need a fresh analysis. There is also no continuous transition theorem joining $p = 1$ to $p \geq 2$; the family considered here has integer p .

Sharper marked approximations. The tilted reference removes the leading location perturbation and leaves a second Charlier correction. Higher signed Charlier corrections are available algebraically. Turning them into positive probability approximants with explicit distance bounds, or proving finer moderate and large deviation estimates, would extend the present central laws.

Broader graph statistics. The inactive count determines the numbers of active and total vertices through affine relations, but does not describe components, cycles, indegrees, or repeated targets. Those observables require additional marks in the directed-graph model. The exact size statistic and label convention should be retained in any such extension.

The results established here are therefore concrete finite-order count, probability, and inverse theorems. Their strengths are exact reduction, global domination, explicit rational generators, and controlled error transfer. Their scope does not depend on an unsupported claim about exponentially small sectors or historical novelty.

[write] **Added 7 October 2026, under Vladimir's standing rule of 4 October 2026.** Every open question of this Part is in this section, with what is missing; the critical marked law is answered in Part II within the conditioned binomial family (note above). The write adds from the non-claims interval-certified values for the tables and diagnostics of Section 7.2, which are floating computations. No claim of the source was found false; the OEIS refinements' first corrections are refuted (Remark 7.1).

A Coefficient and discrepancy reference

For quick comparison, “observed minus proved” means the coefficient in the normalized observed multiplier of Section 7.4 minus the corresponding coefficient in this article. The parameter c is always the one belonging to the indicated p . For $p = 1$ the sole displayed difference is

$$[t] : \quad \frac{1}{8c}.$$

For $p = 2$ the differences through t^3 are

$$\begin{aligned} [t] &: \quad \frac{1}{8c}, \\ [t^2] &: \quad \frac{67}{384c^2} - \frac{13c}{48}, \\ [t^3] &: \quad \frac{3929}{2304} + \frac{497}{3072c^3} + \frac{169c^3}{576}. \end{aligned}$$

For $p = 3$ the differences through t^4 are

$$\begin{aligned} [t] &: \quad \frac{1}{8c}, \\ [t^2] &: \quad \frac{67}{384c^2}, \\ [t^3] &: \quad \frac{497}{3072c^3} - \frac{37c}{64}, \\ [t^4] &: \quad \frac{10721}{3072} + \frac{218831}{1474560c^4}. \end{aligned}$$

These comparisons are about intended refinements, not the validity of leading equivalence under the literal symbol $\sim$.

B Reproduction guide

The archive contains this PDF, editable TeX sources, the exact-arithmetic code, optional high-precision diagnostics, checked outputs, a source guide, and a SHA-256 manifest. No network access is needed for reproduction.

From an extracted archive, the standard-library exact checks are run with

```
python3 -B code/selftest.py
python3 -B -O code/selftest.py
python3 -B code/endpoint.py coefficients --p 2 --order 4
python3 -B code/endpoint.py coefficients --p 3 --order 6
python3 -B code/endpoint.py critical --order 4
python3 -B code/endpoint.py marked --p 3 --order 3 --cumulant 2
```

The core never infers coefficients by fitting numerical values. The rational-polynomial results can also be requested as JSON. The ordinary selftest itself launches a separate optimized-Python guard

check; invoking the whole test under `-O` additionally checks that its correctness tests do not depend on removable assertions.

With the optional dependency in `code/requirements-optional.txt`, run

```
python3 -B code/selftest.py --numerical
python3 -B code/diagnostics.py suite --deep
```

The deep suite reproduces the complete $n = 10000$ sums and the marked $n = 10^9$ diagnostics, together with finite Newton experiments. Its JSON explicitly labels all transcendental values uncertified and records the $p = 3$ preasymptotic warning.

To verify the frozen file inventory and rebuild the document, use

```
python3 -B build.py --verify-only
python3 -B build.py --output ../rebuilt
python3 -B -O build.py --output ../rebuilt-optimized --numerical
```

A PDF build needs `pdflatex` with the standard packages listed in `article.tex`. The output directory must lie outside the extracted package, preserving its frozen inventory. The build runs exact checks, compiles in a disposable directory, and writes a PDF, deterministic ZIP, and check summary to the chosen output directory. With `-numerical`, it also reproduces and compares the supplied high-precision diagnostics.

Deterministic timestamps and archive metadata are used. Byte-identical PDF rebuilding is checked on the documented TeX toolchain; another TeX release can produce different PDF bytes even when the mathematical content is identical. The source manifest still verifies the original frozen files independently. The build distinguishes content verification from claims about universal cross-toolchain PDF identity.

Useful exact recurrences

The public code uses only elementary rational polynomial operations. For the power sums $S_j(r) = \sum_{h=0}^{r-1} h^j$, telescoping $r^{j+1} = \sum_{h=0}^{r-1} ((h+1)^{j+1} - h^{j+1})$ gives

$$S_j(r) = \frac{1}{j+1} \left(r^{j+1} - \sum_{i=0}^{j-1} \binom{j+1}{i} S_i(r) \right).$$

The Touchard recurrence is $T_{d+1}(z) = z(T_d(z) + T'_d(z))$, with $T_0 = 1$. Together with (3.6), (3.7), (5.25), and (4.13), these determine every displayed coefficient.

The public implementation caps resources to keep accidental requests from creating unbounded computations. Extending those caps is a programming choice; it is not evidence of an asymptotic theorem with growing order.

[write] **The shipped files, 7 October 2026.** In ProveIt the PDF and `MANIFEST.sha256` are not shipped; the programs are `code/230-endpoint-*.py` and their recorded outputs and requirements are `data/230-endpoint-*`. The programs import each other under their delivered names, so the report README gives commands that rerun them in a scratch copy with the prefix removed; the builder expects the delivered layout and runs only in a re-extracted archive.

Part II

Optimal positive binomial approximation for the critical inactive count

Bundle Report 235, archive `Report235.zip`, delivered as `article.tex` with ten section files (23 pages). Delivered title block “Report 235 / Optimal Positive Binomial Approximation for the Critical Inactive Count / A global total variation theorem connected to OEIS A360592 / 5 October 2026”. Printed as delivered, with `[write]` notes, except that three unlabelled proofs that repeat Part I are printed once there (a note stands at each place). Report 235’s Section k is Section $k + 8$ here, its equation $(k.j)$ is $(k + 8.j)$ and its Theorem $k.j$ is Theorem $k + 8.j$; its appendix keeps the letter A. Its labels carry `spb:ml:`. “Report 230” in this Part is Part I.

Abstract of this Part (as delivered). Choose a term of $a(n) = \sum_k (n - k)^k \binom{n-k}{k}$ with probability proportional to its weight, and set $R_n = n - 2k$. We determine the globally asymptotically best approximation of R_n by a binomial distribution conditioned to the parity of n , with arbitrary integer trial count and arbitrary success probability whenever conditioning is defined. If $\mu \sim \sqrt{en/2}$ is the explicit tilted Poisson parameter, the optimal total variation error is

$$\inf_B d_{\text{TV}}(\mathcal{L}(R_n), B) \sim \frac{5 \log 2}{8} \sqrt{\frac{2}{\pi}} \frac{\mu^{3/2}}{n^2}.$$

An explicit positive construction attains this bound. Every asymptotically optimal parameter sequence has a uniquely determined mean adjustment at scale n^{-1} and trial count localized to $o(n^{1/4})$. The proof supplies all fixed order marked expansions, a polynomially weighted common reference density comparison, support and rounding estimates, and a localization argument covering rare parity events. Its fine step uses an exact characteristic function Jacobian rather than inferring moment closeness from total variation. The classical Charlier, Hermite, and binomial fitting mechanisms are credited; the contribution is the complete model specific argument and its constants. Computation corroborates the proof and carries separate limitations.

9 The law and the global optimum

The ordinary generating function recorded for OEIS A360592 is

$$\sum_{n \geq 0} a(n) x^n = \sum_{m \geq 0} x^m (1 + mx)^m, \quad a(n) = \sum_{k=0}^{\lfloor n/2 \rfloor} (n - k)^k \binom{n-k}{k}. \quad (9.1)$$

These are formal series; no analytic convergence near the origin is required. Here is a direct interpretation of the weights. On the fixed vertex set $[m]$, choose k active vertices and give each active vertex one image in $[m]$. Loops and repeated images are allowed. There are $\binom{m}{k} m^k$ such partial functions. Give an inactive vertex size one and an active vertex size two, so total size is $m + k = n$. The inactive count is $m - k = n - 2k$. Thus a uniformly chosen object of size n gives

$$\mathbb{P}(R_n = r) = \frac{W_n(r)}{a(n)}, \quad W_n(r) = \left(\frac{n+r}{2} \right)^{(n-r)/2} \binom{(n+r)/2}{(n-r)/2}, \quad (9.2)$$

for $0 \leq r \leq n$, $r \equiv n \pmod{2}$, and zero otherwise. The vertex set is fixed for each m ; no additional labeling factor is inserted. This interpretation is derived here from the entry's generating function, not quoted as an OEIS combinatorial claim. We use $0^0 = 1$ for the empty object.

Throughout the article, $n \rightarrow \infty$ through both parities, and

$$t = n^{-1/2}, \quad c = \sqrt{e/2}, \quad \lambda = c/t, \quad A = \frac{3}{4}, \quad b = -\frac{1}{4}, \quad a = n \bmod 2, \quad \gamma = \frac{5}{8}. \quad (9.3)$$

Let $Q_{\eta,a}$ be $\text{Pois}(\eta)$ conditioned to parity a . Let $\mu > 0$ solve

$$\log(\mu/\lambda) = -\frac{2A\mu + b}{n}, \quad \mu = \frac{n}{2A} W\left(\frac{2A\lambda}{n} e^{-b/n}\right), \quad (9.4)$$

where W is the positive real Lambert function. The left side after moving all terms to the left is strictly increasing from $-\infty$ to $+\infty$, so the solution exists uniquely. Expansion of this equation gives $\mu = \lambda - \frac{3}{2}c^2 + O(t)$.

For $M \in \mathbb{N}$ and $0 < \nu < M$, write $B_{M,\nu,a}$ for $\text{Bin}(M, \nu/M)$ conditioned to parity a . The parameter ν is its *unconditioned* mean. Define the following explicit polynomials:

$$\nu_0 = \lambda - \frac{3}{2}c^2 + \left(\frac{c}{4} + \frac{33c^3}{8}\right)t + \left(\frac{15c^2}{8} - \frac{44c^4}{3}\right)t^2, \quad (9.5)$$

$$\nu_+ = \lambda - 3c^2 + \left(\frac{c}{4} + \frac{99c^3}{8}\right)t + \left(\frac{15c^2}{4} - \frac{176c^4}{3}\right)t^2. \quad (9.6)$$

Set

$$M_0 = \text{round}\left(\frac{\nu_0^2}{\nu_0 - \nu_+}\right), \quad \hat{\nu} = \nu_0 + \gamma(2 \log 2 - 3)\frac{\mu^2}{n^2}, \quad \varepsilon_n = \frac{\mu^{3/2}}{n^2}. \quad (9.7)$$

Any nearest integer convention is allowed. All these probability distributions are well defined for sufficiently large n : $\nu_0 - \nu_+ \rightarrow \frac{3}{2}c^2 > 0$, $M_0 \sim 2n/3$, and $0 < \hat{\nu} < M_0$. Formulas are asymptotic constructions, not claims of admissibility at every small n .

Theorem 9.1 (Global asymptotic optimum). *Let $\mathcal{B}_a$ contain every $\text{Bin}(M, q)$ conditioned to parity a , for $M \in \{0, 1, 2, \dots\}$ and $q \in [0, 1]$, whenever the conditioning event has positive probability. Then*

$$\inf_{B \in \mathcal{B}_a} d_{\text{TV}}(\mathcal{L}(R_n), B) \sim \gamma \log 2 \sqrt{2/\pi} \varepsilon_n. \quad (9.8)$$

The law $B_{M_0, \hat{\nu}, a}$ attains this equivalent. Every sequence with error $O(\varepsilon_n)$ has

$$\nu - \nu_0 = O(\mu^2/n^2), \quad M - M_0 = O(\sqrt{\mu}), \quad (9.9)$$

where $\nu = Mq$. A sequence is asymptotically optimal if and only if

$$\nu - \nu_0 = \gamma(2 \log 2 - 3)\frac{\mu^2}{n^2} + o(\mu^2/n^2), \quad M - M_0 = o(\sqrt{\mu}). \quad (9.10)$$

For the sufficient direction, the laws must be defined; these parameter relations already ensure their eventual interior admissibility.

Here $d_{\text{TV}}(P, Q) = \frac{1}{2} \sum_r |P\{r\} - Q\{r\}|$. The infimum is global within the stated conditioned binomial family. Neither an exact finite n minimizer nor optimality over all positive families is asserted.

The moment matched baseline has the larger equivalent

$$d_{\text{TV}}(\mathcal{L}(R_n), B_{M_0, \nu_0, a}) \sim \frac{5}{16} \{2\phi(0) + 8\phi(\sqrt{3})\} \varepsilon_n, \quad (9.11)$$

where $\phi(z) = (2\pi)^{-1/2} e^{-z^2/2}$. The tuned constant is about 0.34566, compared with 0.47188 in (9.11); their ratio is about 0.73251. The improvement changes the mean by order n^{-1} , despite both means agreeing with the true mean at leading orders.

Relation to the scalar predecessor. Report 230 [6], *Corrected Poisson Endpoint Expansions*, established the scalar fixed order expansion of $a(n)$ at this critical endpoint and the broader fixed parameter family. It explicitly left the critical marked law outside its scope. We credit that scalar foundation and reprove all inputs needed here. The focus of the present article is the full inactive distribution and global positive approximation. All required inputs to Theorem 9.1 are established below.

[write] **7 October 2026.** Report 230 is Part I of this report. Its Scope paragraph indeed asserts no critical marked-law theorem, and its Section 5 assumes $p \geq 2$. The inputs this Part reproves are Part I's at $p = 1$: the reduction and bound of Lemma 10.1 are (2.3) and Lemma 4.2, Lemma 10.2 is (4.10)–(4.11), and the scalar coefficients (11.5) are (4.2) and (4.3).

10 Exact deformation and uniform tail control

Lemma 10.1 (Exact Poisson reduction). *For each permitted r in (9.2),*

$$W_n(r) = (n/2)^{n/2} \frac{\lambda^r}{r!} e^{\Delta_n(r)}, \quad (10.1)$$

$$\Delta_n(r) = \frac{n+r}{2} \log(1+r/n) - \frac{r}{2} + \sum_{h=0}^{r-1} \log\left(1 - \frac{2h}{n+r}\right). \quad (10.2)$$

Furthermore, on the entire finite support,

$$\Delta_n(r) \leq \frac{1 - (r-1)^2}{4n} \leq \frac{1}{4n}. \quad (10.3)$$

Proof. Put $m = (n+r)/2$, $k = (n-r)/2 = m-r$. Then $\binom{m}{k} = m(m-1) \cdots (m-r+1)/r!$. Extracting m^r and multiplying by m^k gives $m^{(n+r)/2}/r!$ times the product in (10.2). Since $\lambda = \sqrt{en}/2$, all remaining powers cancel to give (10.1).

[write] **Printed once, 7 October 2026.** The rest of Report 235's proof, the bound (10.3), is Part I's proof of Lemma 4.2 (the same convexity bound $F(r) \leq r^2/(4n)$ and the same bound $\log(1-x) \leq -x$ for the sum), together with the remarks that every logarithm argument is positive at $r = n$ and that $\Delta_n(1) > 0$; it is printed there.

□

Lemma 10.2 (Finite defect expansion). *For $j \geq 1$, define the rational polynomial*

$$D_j(r) = (-1)^{j+1} \left\{ \frac{r^{j+1}}{2j(j+1)} + \frac{1}{j} \sum_{h=0}^{r-1} ((r-2h)^j - r^j) \right\}. \quad (10.4)$$

The power sum means its polynomial continuation. Its degree is at most $j+1$, and for each fixed L , uniformly for $0 \leq r \leq C\sqrt{n}$ with fixed C ,

$$\Delta_n(r) = \sum_{j=1}^L \frac{D_j(r)}{n^j} + O_L(r^{L+2}/n^{L+1}). \quad (10.5)$$

The first three polynomials are

$$D_1 = -\frac{3}{4}r^2 + r, \quad D_2 = \frac{1}{4}r^3 - \frac{r}{3}, \quad D_3 = -\frac{7}{24}r^4 + \frac{1}{3}r^3. \quad (10.6)$$

[write] **Printed once, 7 October 2026.** Report 235's proof of this lemma is Part I's verification of (4.11), printed there: the defect is rewritten as $\frac{n-r}{2} \log(1+r/n) - \frac{r}{2} + \sum_{h < r} \log(1+(r-2h)/n)$, the first logarithm is expanded through degree $L+1$ and each summed one through degree L , and the remainders give $O_L(r^{L+2}/n^{L+1})$; the range $0 \leq r \leq C\sqrt{n}$ is covered because the logarithm arguments stay within $O(n^{-1/2})$ of one. Direct evaluation gives (10.6), the first three entries of Part I's list after (4.10).

Here and below, constants may depend on fixed orders or on a compact positive interval for the marking parameter. They are uniform in $a \in \{0, 1\}$. We use the following elementary probabilistic facts, supplying details to fix all tail and parity inputs.

Lemma 10.3 (Poisson estimates with parity). *For $X \sim \text{Pois}(\eta)$ with $\eta \rightarrow \infty$, every fixed real $s \geq 0$ satisfies*

$$\mathbb{E}|X - \eta|^s = O_s(\eta^{s/2}).$$

The same bound holds after conditioning on either parity. Polynomially weighted tails outside $|X - \eta| \leq \eta^{3/4}$ are smaller than every inverse power of η . Under $Q_{\eta,a}$,

$$(X - \eta)/\sqrt{\eta} \implies N(0, 1),$$

with convergence of every fixed absolute polynomial moment.

Proof. From $\mathbb{E}e^{z(X-\eta)} = \exp(\eta(e^z - 1 - z))$, the d th centered moment is

$$\mathbb{E}(X - \eta)^d = \sum_{k=0}^{\lfloor d/2 \rfloor} B_{d,k}^{\geq 2} \eta^k, \quad (10.7)$$

where $B_{d,k}^{\geq 2}$ counts partitions into k blocks of size at least two. Its recurrence is

$$B_{d,k}^{\geq 2} = kB_{d-1,k}^{\geq 2} + (d-1)B_{d-2,k-1}^{\geq 2}, \quad B_{0,0}^{\geq 2} = 1,$$

with inadmissible entries zero. This follows either by inserting the last element into an existing block or making its block a pair; equivalently it follows by differentiating the exponential generating function. Even moments and Hölder's inequality prove the real absolute moment bound.

The exponential Markov inequality, optimized at $z = \log(1 + x/\eta)$ for the upper tail and its negative counterpart for the lower tail, gives

$$\mathbb{P}(|X - \eta| \geq x) \leq 2 \exp\{-x^2/(2(\eta + x))\}. \quad (10.8)$$

At $x = \eta^{3/4}$ this is $O(e^{-d\sqrt{\eta}})$. Cauchy–Schwarz with higher even moments proves every weighted version. Conditioning costs a bounded factor because $\mathbb{P}(X \equiv a) = \frac{1}{2}(1 + (-1)^a e^{-2\eta})$.

The conditional characteristic function at $z/\sqrt{\eta}$, centered at η , is

$$e^{-iz\sqrt{\eta}} \frac{e^{\eta(e^{iz/\sqrt{\eta}}-1)} + (-1)^a e^{\eta(-e^{iz/\sqrt{\eta}}-1)}}{1 + (-1)^a e^{-2\eta}}.$$

The first term tends to $e^{-z^2/2}$; the second is exponentially small for bounded z . The characteristic function criterion proves the limit. Higher even moment bounds give uniform integrability for each requested absolute polynomial moment. $\square$

[write] **7 October 2026.** Kept because later labels depend on it. Its parts are Part I's (4.14) and (4.13) (centered moments), (4.9) (the tail at $\eta^{3/4}$), and the conditional central limit theorem used in the proof of Theorem 5.1, here written out for parity.

For $X \sim Q_{\lambda,a}$, (10.5) on its central window implies $\Delta_n(X) \rightarrow -Ac^2$ in probability. The global bound (10.3), together with Lemma 10.3, therefore gives

$$Z_n := \mathbb{E}_{Q_{\lambda,a}}[e^{\Delta_n(X)} \mathbf{1}_{X \leq n}] \longrightarrow e^{-Ac^2} > 0. \quad (10.9)$$

This is also uniform for c in a fixed compact positive interval. In particular, every polynomially weighted tail of the exact law is bounded by a fixed multiple of the corresponding $Q_{\lambda,a}$ tail. Since $\mu - \lambda = O(1)$, the window

$$\mathcal{W}_n = \{r : |r - \mu| \leq \mu^{3/4}\} \quad (10.10)$$

has negligible tails for both the exact law and $Q_{\mu,a}$, with any fixed polynomial weight. It lies below n for large n . This proves the finite support control used throughout.

11 Marked expansions and the tilted density

11.1 A finite scalar and cumulant foundation

To handle marking rigorously, temporarily let c vary over a compact subset of $(0, \infty)$ and set $\lambda = c/t$, keeping Δ_n unchanged. Define

$$F_n(c) = \sum_{\substack{0 \leq r \leq n \\ r \equiv a \pmod{2}}} \frac{(c/t)^r}{r!} e^{\Delta_n(r)}. \quad (11.1)$$

At the physical value $c = \sqrt{e/2}$, $(n/2)^{n/2} F_n(c) = a(n)$.

Proposition 11.1 (Scalar expansion with a finite generator). *For every fixed integer $K \geq 0$, there are computable polynomials $C_j(c) \in \mathbb{Q}[c]$ such that*

$$F_n(c) = \frac{1}{2}e^{c/t - Ac^2} \left(\sum_{j=0}^K C_j(c)t^j + O_K(t^{K+1}) \right), \quad C_0 = 1. \quad (11.2)$$

Give t weight two and u weight one. Form

$$V_K^{\text{sc}} = Ac^2 + \sum_{j=1}^{K+1} t^{2j} D_j((c+u)/t), \quad P_K^{\text{sc}} = [e^{V_K^{\text{sc}}}]_{\text{wt} \leq 2K+1}.$$

Replace every u^d by

$$\sum_{k=0}^{\lfloor d/2 \rfloor} B_{d,k}^{\geq 2} c^k t^{d-k}, \quad (11.3)$$

and retain powers through t^K . This finite algorithm gives the coefficients in (11.2).

Proof. Center now at λ : $U = t(X - \lambda) = tX - c$. The degree bound on D_j implies that V_K^{sc} has no negative powers of t . At $t = 0$ it equals $-2Acu - Au^2$, so every monomial has positive weight. The exponential truncation needs at most powers zero through $2K + 1$.

On $|X - \lambda| \leq \lambda^{3/4}$, Lemma 10.2 with $L = K + 1$ has remainder $O(t^{K+1})$, and $U = O(t^{1/4})$. Both exponents remain bounded. Apply ordinary multivariate Taylor's theorem to $\exp(V_K^{\text{sc}}(s^2, u))$, with $s = \sqrt{t}$, through total degree $2K + 1$. Its remainder is bounded by $C_K(\sqrt{t} + |u|)^{2K+2}$. Formula (11.3), which is (10.7) after scaling, makes the averaged absolute remainder $O(t^{K+1})$. Lemma 10.3 and (10.3) remove all central tails, including the cutoff at n .

For every polynomial f , the exact parity filter is

$$2\mathbb{E}[f(X)\mathbf{1}_{X \equiv a}] = \mathbb{E}f(X) + (-1)^a \mathbb{E}[(-1)^X f(X)]. \quad (11.4)$$

If T_d denotes the d th Touchard polynomial, direct differentiation of $\mathbb{E}z^X = e^{\lambda(z-1)}$ yields $\mathbb{E}[(-1)^X X^d] = e^{-2\lambda} T_d(-\lambda)$. Hence a fixed polynomial in t and U has exponentially small parity correction. Replacing by the full ordinary expectation proves (11.2). Every exponent $i + d - k$ produced from $t^i u^d$ is integral. This also proves the absence of odd quarter powers in the scalar expansion. $\square$

[write] 7 October 2026. At the physical value of c this is Part I's Theorem 4.1 with the same finite generator; the uniformity over a compact range of c , which the marking argument needs, is the extension. The parity filter (11.4) is the display in Part I's proof of that theorem.

The first coefficients and logarithmic coefficients, obtained by these finite substitutions, are

$$C_1 = \frac{c}{4} + \frac{11c^3}{8}, \quad C_2 = \frac{31c^2}{32} - \frac{319c^4}{96} + \frac{121c^6}{128}, \quad (11.5)$$

$$L_1 = \frac{c}{4} + \frac{11c^3}{8}, \quad L_2 = C_2 - C_1^2/2 = \frac{15c^2}{16} - \frac{11c^4}{3}, \quad (11.6)$$

where $\log(\sum C_j t^j) = \sum_{j \geq 1} L_j t^j$.

[write] Printed once, 7 October 2026. Report 235 adds here the direct check of C_1 without the polynomial algorithm ($f(z) = e^{-3z^2/4}$, $\mathbb{E}U^2 = ct$, giving $c + c^3/4 + (c/2)(-3/2 + 9c^2/4) = c/4 + 11c^3/8$); it is printed once, as Part I's Section 4.5.

Theorem 11.2 (All fixed order marked cumulants). *Let $\Theta = c d/dc$. For every fixed $h \geq 1$, $K \geq 0$, the h th cumulant of the law defined by (11.1) satisfies*

$$\kappa_h = \lambda - A 2^h c^2 + \sum_{j=1}^K \Theta^h L_j(c) t^j + O_{h,K}(t^{K+1}). \quad (11.7)$$

In particular, at the physical value of c ,

$$\mathbb{E} R_n = \nu_0 + O(t^3), \quad \text{Var } R_n = v_+ + O(t^3). \quad (11.8)$$

Proof. We do not differentiate the remainder in (11.2). Instead construct the scalar centered polynomial at order $J = K + h$, multiply it by each required raw power r^d , $d \leq h$, and average. On the central window $r^d = t^{-d}(c + U)^d$. The proof of Proposition 11.1, with additional fixed weights, gives raw moment error $O(t^{J+1-d})$, including division by the positive normalizer. The same argument controls all tails. In a moment to cumulant monomial of total degree h , the other raw moment factors have size at most $O(t^{-(h-d)})$. Thus replacing any one raw moment by its error costs $O(t^{J+1-h}) = O(t^{K+1})$; terms with multiple errors are smaller.

It remains to identify the resulting finite polynomial coefficients. Differentiation of the exact finite sum F_n inserts r , and the polynomial identity for the ordinary Poisson expectation is

$$\Theta \mathbb{E} P(c, t, tX - c) = \mathbb{E} [(c\partial_c - c\partial_u + u/t)P(c, t, u)]_{u=tX-c}. \quad (11.9)$$

It follows by differentiating the Poisson weights and the argument $u = tX - c$; the weight derivative is $X - \lambda = u/t$. Each operation lowers weighted degree by at most one. Hence each of the h desired formal differentiations is determined by a finite longer truncation, and agrees with the raw moment calculation whose error was already proved. This identifies its coefficients with $\Theta^h \log F_n$ and proves (11.7). Applying Θ and Θ^2 to (11.6) gives exactly (9.5), (9.6). $\square$

11.2 All fixed order weighted density expansion

Return to the physical c , and let $g_n(r) = \mathbb{P}(R_n = r)/Q_{\mu,a}\{r\}$, with $g_n = 0$ for $r > n$. Define the first three monic Charlier polynomials by

$$\begin{aligned} C_1(r; \mu) &= r - \mu, \\ C_2(r; \mu) &= (r - \mu)^2 - r, \\ C_3(r; \mu) &= (r - \mu)^3 - 3(r - \mu)^2 + (2 - 3\mu)(r - \mu) + 2\mu. \end{aligned} \quad (11.10)$$

They are distinct from the scalar coefficients $C_j(c)$; the arguments distinguish the notation. Put $\beta_0 = \log(\mu/\lambda)$, $\theta = t\mu$ and $u = t(r - \mu)$. The exact cancellation underlying the tilt is

$$D_1(r)/n - \beta_0 r = A\mu^2/n - AC_2(r; \mu)/n. \quad (11.11)$$

Thus $g_n = e^{V_n}/T_n$ on its support, where

$$\begin{aligned} V_n(r) &= \Delta_n(r) - \beta_0 r - A\mu^2/n, \\ T_n &= e^{\lambda - \mu - A\mu^2/n} \frac{1 + (-1)^a e^{-2\lambda}}{1 + (-1)^a e^{-2\mu}} Z_n \longrightarrow 1. \end{aligned} \quad (11.12)$$

The limit follows from $\lambda - \mu \rightarrow 2Ac^2$, $A\mu^2/n \rightarrow Ac^2$, and (10.9). This also controls the unnormalized weighted tails of e^{V_n} .

Theorem 11.3 (Weighted polynomial density generator). *For $K \geq 0$, define*

$$\begin{aligned} V_K(t, u; \theta) &= \sum_{j=1}^{K+1} t^{2j} D_j((\theta + u)/t) + (2A\theta + bt)(\theta + u) - A\theta^2, \\ P_K(t, u; \theta) &= [e^{V_K(t, u; \theta)}]_{\text{wt} \leq 2K+1}, \\ Z_{K,a} &= \mathbb{E}_{Q_{\mu,a}} P_K(t, t(X - \mu); \theta). \end{aligned}$$

Then $Z_{K,a} = 1 + O(t)$ and, for every fixed real $s \geq 0$,

$$\mathbb{E}_{Q_{\mu,a}} \left[|X - \mu|^s \left| g_n(X) - \frac{P_K(t, t(X - \mu); \theta)}{Z_{K,a}} \right| \right] = O_{K,s}(\mu^{s/2} t^{K+1}). \quad (11.13)$$

The constants are uniform for θ in a fixed compact positive interval. At $s = 0$ the weight is interpreted as one.

Proof. By (11.11), all monomials of V_K have weighted degree at least two: its first defect contribution is $-Au^2 + At(\theta + u)$, and all later defects have that property too. Consequently powers zero through K suffice for the exponential truncation.

On (10.10), Lemma 10.2 gives exponent error $O(t^{K+1})$. The exponents are bounded there. Taylor's theorem in $(\sqrt{t}, u)$ gives remainder $O((\sqrt{t} + |u|)^{2K+2})$. With $U = t(X - \mu)$, Lemma 10.3 gives $\mathbb{E}|U|^d = O_d(t^{d/2})$ for every fixed real $d \geq 0$. Multiply the Taylor error by $|X - \mu|^s = t^{-s}|U|^s$ and average. The result is $O(t^{K+1-s/2}) = O(\mu^{s/2} t^{K+1})$. The exponent error has the same weighted bound. All polynomial and exact density tails are negligible by (11.12) and Section 10; this includes $r > n$.

The scalar expectation is $1 + O(t)$. Its exact error is $O(t^{K+1})$, while the polynomial's weighted absolute moment is $O(\mu^{s/2})$. Dividing by these bounded away from zero normalizers preserves the weighted error. Formula (11.4) and the centered moment polynomial compute $Z_{K,a}$ exactly, with only exponentially small deviations from the ordinary Poisson expectation. $\square$

For $K = 1$, direct substitution gives

$$P_1 = 1 - Au^2 + t(A\theta + \theta^3/4) + t(A + 3\theta^2/4)u, \quad Z_{1,a} = 1 + \theta^3 t/4 + O(e^{-d\mu}).$$

Expanding its quotient proves the useful refinement

$$g_n = 1 - \frac{A}{n}C_2 + \frac{3\mu^2}{4n^2}C_1 + \eta_n, \quad \mathbb{E}_Q[|X - \mu|^s |\eta_n|] = O_s(\mu^{s/2} t^2). \quad (11.14)$$

Here and in the sequel $Q = Q_{\mu,a}$. These polynomial approximants are signed; their nonnegativity at remote lattice points is neither required nor claimed.

Corollary 11.4 (Sharp Poisson comparisons). *Uniformly in parity,*

$$d_{\text{TV}}(\mathcal{L}(R_n), Q_{\lambda,a}) \sim A\sqrt{2/\pi} \lambda^{3/2}/n, \quad (11.15)$$

$$d_{\text{TV}}(\mathcal{L}(R_n), Q_{\mu,a}) \sim A\sqrt{2/(\pi e)} \mu/n. \quad (11.16)$$

In particular $(R_n - \mu)/\sqrt{\mu} \implies N(0, 1)$.

Proof. Lemma 10.3 gives $\mathbb{E}_Q|C_2|/\mu \rightarrow \mathbb{E}|Z^2 - 1| = 4\phi(1)$, where Z is standard normal. Equation (11.14) has smaller remainder $O(\mu^{5/2}/n^2)$ after its linear term is included, proving (11.16). For bounded δ , the exact ratio of conditioned Poisson masses, Taylor expanded on the central window, is

$$\frac{Q_{\lambda+\delta,a}\{r\}}{Q_{\lambda,a}\{r\}} = 1 + \delta(r - \lambda)/\lambda + \rho(r), \quad \mathbb{E}_{Q_{\lambda,a}}|\rho| = O(\lambda^{-1}).$$

Indeed its logarithm is $r \log(1 + \delta/\lambda) - \delta$ plus an exponentially small parity normalizer; quadratic centered moments bound the exponential Taylor remainder. The tails follow from Lemma 10.3. For δ converging to a nonzero constant, the TV equivalent is $|\delta|/\sqrt{2\pi\lambda}$. Take $\delta = \mu - \lambda \rightarrow -2Ac^2$. The triangle and reverse triangle inequalities, using the $O(n^{-1/2})$ error in (11.16), prove (11.15). TV convergence to $Q_{\mu,a}$ transfers its central limit theorem. $\square$

[write] **7 October 2026.** These are the equivalents (5.6) and (5.7), with the same constants, at $p = 1$, where Part I does not assert them. The write recomputed the two ratios for the table of Section 16 (note there).

12 Positive binomial approximants and the cubic remainder

The distributional theorem rests on a weighted density estimate, not on matching finitely many cumulants. We now prove that estimate relative to the *common* reference $Q_{\mu,a}$. This reference has positive mass at every nonnegative point of the required parity, so it accommodates both finite supports.

12.1 Finite logarithmic expansion and centered bounds

For $r \leq M$, the ordinary binomial to Poisson logarithmic ratio is exactly

$$H_{M,\nu}(r) = \sum_{h=0}^{r-1} \log(1 - h/M) + (M - r) \log(1 - \nu/M) + \nu. \quad (12.1)$$

If r/M and ν/M are sufficiently small, finite Taylor expansion gives, for fixed L ,

$$H_{M,\nu}(r) = \sum_{j=1}^L \frac{B_j(r; \nu)}{M^j} + O_L \left(\frac{r^{L+2} + \nu^{L+2} + r\nu^{L+1}}{M^{L+1}} \right), \quad (12.2)$$

$$B_j(r; \nu) = -\frac{1}{j} \sum_{h=0}^{r-1} h^j + \frac{r\nu^j}{j} - \frac{\nu^{j+1}}{j+1}. \quad (12.3)$$

To obtain the stated remainder, expand the first log terms through degree L , and $M \log(1 - \nu/M)$ through degree $L + 1$. Each finite geometric remainder is bounded uniformly on this range. In particular $B_1 = -C_2(r; \nu)/2$ and

$$\begin{aligned} B_2 &= -r^3/6 + r^2/4 - r/12 + r\nu^2/2 - \nu^3/3, \\ B_3 &= -r^4/12 + r^3/6 - r^2/12 + r\nu^3/3 - \nu^4/4. \end{aligned} \quad (12.4)$$

Centering at $y = r - \nu$ is essential. Direct substitution gives

$$B_2 = \nu^2/4 - \nu/12 + (\nu/2 - 1/12)y + (-\nu/2 + 1/4)y^2 - y^3/6, \quad (12.5)$$

$$B_3 = \nu^3/6 - \nu^2/12 + (\nu^2/2 - \nu/6)y + (-\nu^2/2 + \nu/2 - 1/12)y^2 + (-\nu/3 + 1/6)y^3 - y^4/12. \quad (12.6)$$

For $\nu - \mu = O(t)$, Lemma 10.3 now proves

$$\mathbb{E}_Q[|X - \mu|^s |B_2(X; \nu)|] = O_s(\mu^{2+s/2}), \quad \mathbb{E}_Q[|X - \mu|^s |B_3(X; \nu)|] = O_s(\mu^{3+s/2}). \quad (12.7)$$

If $M \asymp n$, the B_3/M^3 term is therefore $O_s(\mu^{s/2}t^3)$. The $L = 3$ remainder of (12.2) has the same bound, since $\mu^5/M^4 = O(t^3)$. An uncentered $O(\mu^4)$ estimate for B_3 would be too weak and is not used.

12.2 Parity, support, and rounding

For the binomial laws constructed here, $M \asymp n$, $\nu \sim \mu$, and $q = \nu/M \rightarrow 0$. Their ordinary exponential moment satisfies

$$\mathbb{E}e^{zX} = (1 + q(e^z - 1))^M \leq e^{\nu(e^z - 1)} \quad (z \in \mathbb{R}),$$

so the upper and lower Chernoff bounds of the Poisson type apply. Fixed polynomial moments have polynomial growth, either by factorial moments or by differentiating this finite generating function. Cauchy–Schwarz then proves negligible weighted tails outside (10.10). Moreover

$$\mathbb{P}(X \equiv a) = \frac{1}{2}\{1 + (-1)^a(1 - 2\nu/M)^M\} = \frac{1}{2} + O(e^{-d\mu}). \quad (12.8)$$

Thus conditioning only changes the log density by an exponentially small constant in all these local constructions and costs a bounded factor in tail estimates. For large n , the common central window lies below both M and n . Its complement includes the entire true mass above M , so no support contribution is discarded.

Rounding M^* by a bounded amount changes $1/M$ by $O(n^{-2})$. In $-C_2/(2M)$ this costs

$$O_s(\mu^{1+s/2}/n^2) = O_s(\mu^{s/2}t^3) \quad (12.9)$$

in weighted $L^1(Q)$. The B_2/M^2 , B_3/M^3 terms and the Taylor remainder are smaller at this precision. The change of the ordinary variance is $O(\nu^2/M^2) = O(t^2)$. Accordingly, moment matching is exact only at the real trial count; it is not claimed exact after rounding.

Theorem 12.1 (Complete weighted cubic comparison). *Let h_0 be the density of $B_{M_0, \nu_0, a}$ relative to $Q = Q_{\mu, a}$, extended by zero for $r > M_0$. For every fixed real $s \geq 0$,*

$$\mathbb{E}_Q \left[|X - \mu|^s \left| g_n(X) - h_0(X) - \frac{\gamma}{n^2} C_3(X; \mu) \right| \right] = O_s(\mu^{s/2}n^{-3/2}). \quad (12.10)$$

The same conclusion holds if the t^2 term in v_+ is omitted when forming M_0 .

Proof. Write $\theta = t\mu$ and $u = t(r - \mu)$. We retain monomials of weighted degree at most five. A monomial $t^i u^j$ of weight at least six has weighted expected absolute value $O_s(\mu^{s/2} t^3)$, by Lemma 10.3. The finite Taylor bounds in Section 11 give the same conclusion for nonpolynomial remainders. All the following log identities are asserted on the common window (10.10), where both densities are positive.

The exact relation between c and θ is

$$c = \theta \exp(3\theta t/2 - t^2/4). \quad (12.11)$$

Substitute this into (9.5), (9.6) to obtain

$$\nu_0 = \mu + \frac{3}{4}\theta^3 t + \left(\frac{21}{8}\theta^2 - \frac{55}{24}\theta^4\right)t^2 + O(t^3), \quad (12.12)$$

$$v_+ = \mu - \frac{3}{2}\theta^2 + \frac{9}{2}\theta^3 t + O(t^2),$$

$$M_0 = \frac{2}{3t^2} + \frac{5\theta}{3t} + O(1), \quad \frac{1}{M_0} = \frac{3}{2}t^2 - \frac{15}{4}\theta t^3 + O(t^4). \quad (12.13)$$

The $O(1)$ in M_0 includes integer rounding. Its effect is covered by (12.9); omitting the t^2 variance term also changes M_0 only by $O(1)$. A mean error $O(t^3)$ changes the density in weighted L^1 by $O_s(\mu^{s/2} t^{7/2})$: the first Poisson likelihood term is the mean increment times $(r - \mu)/\mu$, and its remainder is smaller by the centered moment bounds.

For clarity we give the actual polynomial identities rather than infer them from cumulants. Theorem 11.3 at $K = 2$, or substitution of D_1, D_2, D_3 and the centered moments, gives

$$\log T_n = \frac{\theta^3}{4}t + \left(-\frac{7\theta^4}{24} + \frac{21\theta^2}{16}\right)t^2 + O(t^3). \quad (12.14)$$

One can verify the scalar normalization directly: the ordinary expectation of the weighted exponential through degree five is

$$1 + \frac{\theta^3}{4}t + \left(\frac{\theta^6}{32} - \frac{7\theta^4}{24} + \frac{21\theta^2}{16}\right)t^2 + O(t^3);$$

taking its logarithm cancels $\theta^6/32$. Parity corrections are exponential. Define

$$L_* = -\frac{3}{4}u^2 + \frac{3}{4}\theta t + \left(\frac{3}{4} + \frac{3}{4}\theta^2\right)tu + \frac{3}{4}\theta tu^2 - \frac{21}{16}\theta^2 t^2 - \frac{7}{6}\theta^3 t^2 u. \quad (12.15)$$

Equations (10.6), (11.11), and (12.14) yield

$$\log g_n = L_* + \frac{1}{4}tu^3 + \mathcal{R}_g. \quad (12.16)$$

For the binomial density add the Poisson mean shift $r \log(\nu_0/\mu) - (\nu_0 - \mu)$ to (12.1). Use (12.12), (12.13), B_1, B_2 and the controlled B_3 remainder. This gives the second explicit identity

$$\log h_0 = L_* - \frac{3}{8}tu^3 + \frac{15}{8}\theta t^2 u + \mathcal{R}_h. \quad (12.17)$$

Both remainders have weighted L^1 norm $O_s(\mu^{s/2} t^3)$. To see that every omitted term is covered, the defect expansion through D_3 has raw error $O(r^5/n^4)$, hence averaged order $O(t^3)$; its retained lower degree terms and normalizer are controlled by the finite generator. The binomial B_3 and raw remainder were bounded in (12.7). Mean and inverse trial parameter errors were handled above. All remaining finite monomials have weight at least six.

Subtracting (12.17) from (12.16) leaves

$$\frac{5}{8}t(u^3 - 3\theta tu) + \mathcal{R}. \quad (12.18)$$

Exponentiation preserves this degree five difference. Indeed the log densities have no term below weight two; multiplying the retained difference by a positive weight log term starts at weight seven. Finite exponential Taylor remainders on the window have the weighted estimates already proved in Theorem 11.3. They also bound products involving discarded terms, since the exponents are uniformly bounded there. Finally,

$$t^4 C_3 = tu^3 - 3\theta t^2 u - 3t^2 u^2 + 2\theta t^3 + 2t^3 u. \quad (12.19)$$

The last three terms have weight at least six. Combining these identities proves (12.10) on the central window. The weighted tail estimates for g_n, h_0 and the Charlier polynomial extend it to the whole common reference space, including both support cutoffs. $\square$

Lemma 10.3 implies $C_3(X; \mu)/\mu^{3/2} \implies H_3(Z) = Z^3 - 3Z$ with first absolute moment convergence. Splitting the Gaussian integral at $0, \pm\sqrt{3}$ gives

$$\mathbb{E}|H_3(Z)| = 2\phi(0) + 8\phi(\sqrt{3}). \quad (12.20)$$

For example, integrate each polynomial times ϕ using $\phi' = -z\phi$; the antiderivative of $(z^3 - 3z)\phi(z)$ is $-(z^2 - 1)\phi(z)$. Since $n^{-3/2} = o(\varepsilon_n)$, division of (12.10) by two proves (9.11).

12.3 The positive approximation hierarchy

There are useful less accurate positive laws between the two Poisson references and the moment matched binomial. Let $M_s = \text{round}(2n/3)$ and $\nu_1 = \mu + 3\mu^3/(4n^2)$. Then

$$d_{\text{TV}}(\mathcal{L}(R_n), B_{M_s, \mu, a}) \sim \frac{3}{8}\sqrt{2/\pi} \mu^{5/2}/n^2, \quad (12.21)$$

$$d_{\text{TV}}(\mathcal{L}(R_n), B_{M_s, \nu_1, a}) \sim \frac{15}{4}\phi(1)\mu^2/n^2. \quad (12.22)$$

Here is a coefficient proof of both assertions. The centered binomial expansion gives $h_s = 1 - AC_2/n + O_{L^1}(t^2)$. Comparison with (11.14) leaves $3\mu^2 C_1/(4n^2)$; its absolute expectation is asymptotic to $\sqrt{2/\pi}\mu^{1/2}$. This proves (12.21).

For the mean corrected law, the first mean shift adds $(3/4)\theta^2 tu$, canceling the unmatched weight three term. The weight at most four log polynomials are, respectively,

$$\begin{aligned} \log g_n &= -\frac{3}{4}u^2 + \frac{3}{4}\theta t + \left(\frac{3}{4} + \frac{3}{4}\theta^2\right)tu + \frac{3}{4}\theta tu^2 - \frac{21}{16}\theta^2 t^2 + \dots, \\ \log h_{s,1} &= -\frac{3}{4}u^2 + \frac{3}{4}\theta t + \left(\frac{3}{4} + \frac{3}{4}\theta^2\right)tu - \frac{9}{8}\theta tu^2 + \frac{9}{16}\theta^2 t^2 + \dots. \end{aligned}$$

Their difference is $(15/8)\theta t(u^2 - \theta t)$, which equals $(15/8)\mu C_2/n^2$ modulo weight at least five. Products in exponentiation have still higher weight. The same Taylor and tail estimates prove an $O_{L^1}(t^{5/2})$ remainder. Since $\mathbb{E}_Q|C_2| \sim 4\phi(1)\mu$, (12.22) follows.

Thus the sharp error orders, in succession, are $n^{-1/4}$ for $Q_{\lambda,a}$, $n^{-1/2}$ for $Q_{\mu,a}$, $n^{-3/4}$ for the simple binomial, n^{-1} after its first mean correction, and $n^{-5/4}$ after fitting both parameters. The final TV tuning preserves the last rate and reduces its constant. Every reference in this hierarchy is a genuine positive probability law.

13 Uniform local sensitivity and the optimal Gaussian fit

Let α, β range over any fixed bounded subset of $\mathbb{R}^2$. Define

$$\nu(\alpha) = \nu_0 + \gamma\alpha\frac{\mu^2}{n^2}, \quad M(\beta) = \text{round}\left(M_0 + \frac{5}{9}\beta\sqrt{\mu}\right), \quad (13.1)$$

and let $h_{\alpha,\beta}$ be the density of $B_{M(\beta),\nu(\alpha),a}$ relative to Q . The mean and trial variations need not correspond to matching any two exact moments.

Theorem 13.1 (Uniform weighted sensitivity). *For each fixed $s \geq 0$, uniformly on bounded (α, β) sets,*

$$\mathbb{E}_Q\left[|X - \mu|^s \left|g_n - h_{\alpha,\beta} - \frac{\gamma}{n^2}\{C_3 - \alpha\mu C_1 - \beta\sqrt{\mu}C_2\}\right|\right] = O_s(\mu^{s/2}n^{-3/2}). \quad (13.2)$$

All Charlier polynomials here have argument $(X; \mu)$. Consequently

$$\frac{d_{\text{TV}}(\mathcal{L}(R_n), B_{M(\beta),\nu(\alpha),a})}{\varepsilon_n} \longrightarrow \frac{\gamma}{2}J(\alpha, \beta), \quad (13.3)$$

uniformly on compact sets, where

$$J(\alpha, \beta) = \mathbb{E}|H_3(Z) - \alpha H_1(Z) - \beta H_2(Z)|, \quad H_1(z) = z, \quad H_2(z) = z^2 - 1, \quad H_3(z) = z^3 - 3z. \quad (13.4)$$

Proof. Let $\delta = \gamma\alpha\mu^2/n^2 = O(t^2)$ and $D = M(\beta) - M_0 = (5/9)\beta\sqrt{\mu} + O(1)$. In the Poisson mean shift part of the logarithm, replacing ν_0 by $\nu_0 + \delta$ has leading term $\delta(r - \mu)/\mu$. Its weighted expected absolute size is $O_s(\mu^{s/2}t^{5/2})$. Replacing ν_0 by μ in this leading coefficient costs $O_s(\mu^{s/2}t^4)$ or smaller, since $\nu_0 - \mu = O(t)$. Its quadratic log error is bounded by $O_s(\mu^{s/2}\delta^2/\mu)$, of order t^5 after removing the weight. The derivative of $-C_2(r; \nu)/(2M)$ with respect to ν adds $\delta(r - \nu)/M$, of weighted order $t^{7/2}$. The changes in B_2/M^2 and the already bounded higher terms are no larger at the required t^3 precision.

The trial variation in the leading logarithm is

$$\frac{D}{2M_0^2}C_2(r; \mu) = \frac{\gamma\beta\sqrt{\mu}}{n^2}C_2(r; \mu) + O_{L_s^1}(\mu^{s/2}t^3),$$

where $O_{L_s^1}$ denotes the corresponding centered weighted norm. The coefficient follows from $\frac{1}{2}(3/2)^2(5/9) = 5/8$ and $M_0/n = 2/3 + O(t)$. The $O(1)$ rounding part contributes $O_s(\mu^{s/2}\mu/n^2) = O_s(\mu^{s/2}t^3)$. The quadratic trial increment contributes at most

$$O_s(\mu^{s/2}D^2\mu/M_0^3) = O_s(\mu^{s/2}\mu^2/n^3) = O_s(\mu^{s/2}t^4).$$

The change to B_2/M^2 has weighted size at most $O_s(\mu^{s/2}\mu^2M_0^{-2}|D|/M_0) = O_s(\mu^{s/2}t^{7/2})$. The explicit centered polynomials (12.5)–(12.6) give the same or smaller bounds for mean changes, mixed terms, and higher logarithms.

Thus the log increment is

$$\frac{\gamma}{n^2}\{\alpha\mu C_1 + \beta\sqrt{\mu}C_2\} + O_{L_s^1}(\mu^{s/2}t^3).$$

Its retained part has weight five. Products with the existing weight two density correction have weight at least seven, and squares of the increment have weight at least ten. Finite exponential Taylor estimates therefore turn this into the same density increment. The window, tail, support, and parity arguments of Section 12 are uniform for these bounded variations. Subtracting from Theorem 12.1 proves (13.2).

By Lemma 10.3, $C_j/\mu^{j/2}$ converge jointly to $H_j(Z)$ for $j = 1, 2, 3$, with enough even moment control for uniform integrability. For fixed parameters this gives (13.3). Uniformity on compact parameter sets follows from the uniform remainder and the Lipschitz bound

$$|J_n(\alpha, \beta) - J_n(\alpha', \beta')| \leq |\alpha - \alpha'| \mathbb{E}_Q |C_1| / \sqrt{\mu} + |\beta - \beta'| \mathbb{E}_Q |C_2| / \mu,$$

whose right side is uniformly bounded by a fixed multiple of parameter distance. A finite net in each compact set completes the proof. $\square$

Lemma 13.2 (Unique Gaussian minimizer). *The function J in (13.4) has the unique global minimizer*

$$\alpha_* = 2 \log 2 - 3, \quad \beta_* = 0, \quad J(\alpha_*, 0) = 2 \log 2 \sqrt{2/\pi}. \quad (13.5)$$

Proof. Pair z with $-z$. For odd $O(z) = H_3(z) - \alpha H_1(z)$ and even $E(z) = H_2(z)$,

$$\frac{|O - \beta E| + |O + \beta E|}{2} = \max(|O|, |\beta E|) \geq |O|.$$

If $\beta \neq 0$, the inequality is strict on an interval about zero: $O(0) = 0$ and $E(0) = -1$. Thus the unique best β for each α is zero.

At $\beta = 0$, write $J(\alpha, 0) = \mathbb{E}\{|Z| |Z^2 - (3 + \alpha)|\}$. Under the probability measure with density $|z|\phi(z)/\mathbb{E}|Z|$, the variable $W = Z^2$ has density $\frac{1}{2}e^{-w/2}$ on $w > 0$. Its unique median is $m = 2 \log 2$. The absolute deviation function is uniquely minimized at a median: differentiating its finite integral gives $2\mathbb{P}(W \leq q) - 1$ for $q > 0$, while it is strictly decreasing for $q < 0$. Hence $3 + \alpha = m$ uniquely. Finally direct integration gives $\mathbb{E}|W - m| = m - 2 + 4e^{-m/2} = m$. Multiplication by $\mathbb{E}|Z| = \sqrt{2/\pi}$ proves (13.5). $\square$

Theorem 13.1 and Lemma 13.2 prove the claimed tuned construction and its optimality among bounded local parameter perturbations. To reach the global theorem, it remains to prove that every globally competitive binomial belongs to this local class. That step cannot follow from cumulant comparison alone: total variation controls bounded test functions and does not, without additional restrictions, force closeness of unbounded moments.

14 Global localization including rare parity events

We first obtain estimates valid for the *entire* conditioned binomial family, even when the conditioning probability is tiny. They do not assume parameters close to the proposed construction.

Lemma 14.1 (Uniform conditioned binomial bounds). *Let $X \sim \text{Bin}(M, q)$, $\nu = Mq$, and $v = Mq(1 - q)$. For either defined parity conditional law X_a , there is an absolute constant C such that*

$$|\mathbb{E}X_a - \nu| \leq 1, \quad \text{Var } X_a \leq 8(v + 1), \quad \max_k \mathbb{P}(X_a = k) \leq \frac{C}{\sqrt{v + 1}}. \quad (14.1)$$

Proof. If $M \leq 1$ or $q \in \{0, 1\}$, any defined conditioned law is deterministic, and the assertions follow directly. Otherwise put $z = 1 - 2q$, $\rho = |z| < 1$, and $\sigma = (-1)^a$. Differentiating $(1 - q + qs)^M$ at $s = -1$ gives the exact identities

$$\begin{aligned}\mathbb{P}(X \equiv a) &= (1 + \sigma z^M)/2, \\ \mathbb{E}[(X - \nu)(-1)^X] &= -2vz^{M-1}, \\ \mathbb{E}[(X - \nu)^2(-1)^X] &= v(4v - 1)z^{M-2}.\end{aligned}\tag{14.2}$$

These are polynomial identities at $z = 0$, including $M = 2$. In particular,

$$|\mathbb{E}X_a - \nu| \leq \frac{2v\rho^{M-1}}{1 - \rho^M} = \frac{M(1 - \rho^2)\rho^{M-1}}{2(1 - \rho^M)} \leq \frac{1 + \rho}{2} \leq 1,$$

because $1 - \rho^M \geq M(1 - \rho)\rho^{M-1}$. The case $\rho = 0$ is interpreted directly or by continuity.

Since $\log \rho \leq (\rho^2 - 1)/2$, we have $\rho^M \leq e^{-2v}$ and $\rho^{M-2} \leq e^{1-2v}$. From (14.2),

$$\mathbb{E}[(X_a - \nu)^2] \leq \frac{v + v|4v - 1|\rho^{M-2}}{1 - \rho^M}.$$

For $0 < v \leq 1$, use $|4v - 1| \leq 3$, $\rho^{M-2} \leq 1$, and $1 - e^{-2v} \geq (1 - e^{-2})v$ to bound this by $4/(1 - e^{-2}) < 5$. For $v \geq 1$, use $|4v - 1| \leq 4v$ and $4ve^{1-2v} \leq 4/e$ to bound it by $v(1 + 4/e)/(1 - e^{-2}) < 3v$. Variance about the conditional mean is no larger, proving the displayed variance bound with room to spare.

Ordinary Fourier inversion and $|1 - q + qe^{is}|^M \leq \exp(-2v \sin^2(s/2))$ give

$$\max_k \mathbb{P}(X = k) \leq \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-2v \sin^2(s/2)} ds \leq C_0/\sqrt{v}.$$

The last estimate follows from $\sin(|s|/2) \geq |s|/\pi$ for $|s| \leq \pi$ and the Gaussian integral. If $v \geq 1$, conditioning costs at most $2/(1 - e^{-2})$. If $v < 1$, the trivial atom bound one suffices after enlarging C . This proves (14.1) uniformly, including rare parity events. $\square$

Lemma 14.2 (Coarse localization from TV convergence). *If $B_n \in \mathcal{B}_a$ and $d_{\text{TV}}(\mathcal{L}(R_n), B_n) \rightarrow 0$, then their ordinary parameters satisfy*

$$\frac{\nu_n - \mu}{\sqrt{\mu}} \rightarrow 0, \quad \frac{v_n}{\mu} \rightarrow 1, \quad q_n \rightarrow 0.\tag{14.3}$$

Proof. The target has a standard normal limit on the $\sqrt{\mu}$ scale by Corollary 11.4. In particular it has vanishing concentration on any interval of length $o(\sqrt{\mu})$, uniformly in the interval's center. To justify the uniformity, its standardized distribution functions converge uniformly to the continuous normal distribution function: prove this using a finite grid on a compact interval and tightness outside it. The probability of an interval is then bounded by two uniform errors plus the normal modulus of continuity.

If $v_n/\mu \rightarrow \infty$ along a subsequence, the atom bound in (14.1) implies that B_n places vanishing mass in every fixed window $[\mu - K\sqrt{\mu}, \mu + K\sqrt{\mu}]$. This contradicts target normal convergence and small TV. If $v_n/\mu \rightarrow 0$, its conditional variance is $O(v_n + 1)$. Chebyshev's inequality places probability tending to one in an interval of half width $((v_n + 1)\mu)^{1/4} = o(\sqrt{\mu})$ about its conditional mean, again contradicting the target concentration bound. These two exclusions imply v_n/μ is bounded above and bounded away from zero.

The mean displacement in (14.1) and Chebyshev's inequality then imply $(\nu_n - \mu)/\sqrt{\mu}$ is bounded. Indeed, if the centers separate by an unbounded number of $\sqrt{\mu}$ units while conditional variance remains $O(\mu)$, the competitor's mass in every fixed target window tends to zero.

Take an arbitrary convergent subsequence of these standardized coordinates, with limits $x \in \mathbb{R}$ and $y \in (0, \infty)$. Ordinary binomial standardized characteristic functions converge to the normal characteristic function whenever $v \rightarrow \infty$. This can be checked without a uniform moment assumption beyond Bernoulli variables: their centered third absolute moment sum is at most v , so finite Taylor expansion of the product at frequency $u/\sqrt{v}$ gives log characteristic function $-u^2/2 + O_u(v^{-1/2})$.

Parity conditioning preserves this weak limit. Its exact characteristic function is

$$\frac{(1 - q + qe^{is})^M + \sigma(1 - q - qe^{is})^M}{1 + \sigma(1 - 2q)^M}. \quad (14.4)$$

At $s = u/\sqrt{v}$, the second numerator has modulus at most $\exp(-2v \cos^2(s/2))$, whereas $|(1 - 2q)^M| \leq e^{-2v}$. Both disappear as $v \rightarrow \infty$. Thus $(B_n - \mu)/\sqrt{\mu}$ tends along this subsequence to $N(x, y)$. TV convergence forces it also to tend to $N(0, 1)$, so $x = 0, y = 1$. Every subsequential limit is the same, proving the first two assertions. They imply $\nu_n \sim \mu$ and $q_n = 1 - v_n/\nu_n \rightarrow 0$.

This argument does not claim that conditional and unconditional binomials are close in TV; they generally are not. $\square$

Proposition 14.3 (Fine localization by an exact derivative map). *If $d_{\text{TV}}(\mathcal{L}(R_n), B_n) = O(\varepsilon_n)$, then (9.9) holds.*

Proof. Theorem 12.1 gives $d_{\text{TV}}(\mathcal{L}(R_n), B_{M_0, \nu_0, a}) = O(\varepsilon_n)$. By the triangle inequality,

$$d_{\text{TV}}(B_n, B_{M_0, \nu_0, a}) = O(\varepsilon_n). \quad (14.5)$$

Let $v_0 = \nu_0(1 - \nu_0/M_0)$ be the *actual* ordinary variance after rounding; it is not set equal to v_+ . Introduce coordinates

$$x = (\nu - \nu_0)/\sqrt{\mu}, \quad y = (v - v_0)/\mu. \quad (14.6)$$

By Lemma 14.2, $x, y \rightarrow 0$. Fix a real $u \neq 0$, say $u = 1$, and let $\Psi_n(\nu, v)$ be (14.4) at frequency $s = u/\sqrt{\mu}$, centered by ν_0 .

In the local region $\nu > v > 0$, substitute $q = 1 - v/\nu$ and the real trial count $M = \nu^2/(\nu - v) = \nu/q$. Use consistent analytic logarithms to extend this expression smoothly to noninteger M . This extension is only a differentiable formula, not a probability law for real M . The actual endpoints remain integer trial laws. The segment between baseline and competitor lies in the convex region $\nu > v > 0$; along it $q = o(1)$ and the standardized coordinates are uniformly $o(1)$. All log bases are in a fixed neighborhood of the positive real axis for large n .

Put $w = e^{is} - 1$ and

$$f(q) = \frac{\log(1 + qw)}{q}, \quad L(\nu, v) = \nu f(q), \quad \partial_\nu L = f + (1 - q)f', \quad \partial_v L = -f'. \quad (14.7)$$

The singularity at $q = 0$ is removable. Finite power expansion, uniform for small q , gives

$$\begin{aligned} f + (1 - q)f' &= w - w^2/2 + O(|w|^3) = is + O(|s|^3), \\ -f' &= w^2/2 + O(|w|^3) = -s^2/2 + O(|s|^3). \end{aligned} \quad (14.8)$$

The centered first root tends uniformly along the segment to $e^{-u^2/2}$. The other root has exponent $\nu \log(1 - q(1 + e^{is}))/q$, whose real part is at most $-d\nu$ for small q . Its quotient and first q derivative are bounded at $q = 0$. Applying the same exact differentiation rules shows that its first ν, v derivatives are bounded multiples of $e^{-d\nu}$. The parity denominator and its derivatives have analogous estimates. They remain negligible after multiplication by $\sqrt{\mu}$ or μ .

It follows from (14.8) that, uniformly along the segment,

$$\partial_x \Psi_n \rightarrow iue^{-u^2/2}, \quad \partial_y \Psi_n \rightarrow -\frac{1}{2}u^2 e^{-u^2/2}.$$

As a real matrix mapping (x, y) to $(\Re \Psi, \Im \Psi)$, this is

$$J_\infty = e^{-u^2/2} \begin{pmatrix} 0 & -u^2/2 \\ u & 0 \end{pmatrix}, \quad s_{\min}(J_\infty) = e^{-u^2/2} \min(|u|, u^2/2) > 0. \quad (14.9)$$

Integrating the exact derivatives along the segment therefore gives

$$\Psi_n(\nu, v) - \Psi_n(\nu_0, v_0) = J_\infty(x, y) + o(\sqrt{x^2 + y^2}).$$

The same limiting matrix is uniform on the entire segment; hence there is no cancellation of rotating derivative directions. For large n this yields the inverse bound

$$|x| + |y| \leq C_u |\Psi_n(\nu, v) - \Psi_n(\nu_0, v_0)|. \quad (14.10)$$

Crucially, the error is multiplicative in displacement, not an additive central limit error. Bounded characteristic function test values differ by at most twice TV. Equations (14.5) and (14.10) now prove

$$|\nu - \nu_0| = O(\varepsilon_n \sqrt{\mu}), \quad |v - v_0| = O(\varepsilon_n \mu). \quad (14.11)$$

Finally $d_0 = \nu_0 - v_0 = \nu_0^2/M_0 \sim 2A\mu^2/n$, a positive quantity bounded away from zero. The relative change of $d = \nu - v$ is $O(\sqrt{\mu}/n) = o(1)$. In $M = \nu^2/d$, the numerator change is $O(\mu|\nu - \nu_0|)$ and the reciprocal denominator change is $O(|d - d_0|/d_0^2)$. Substitution of (14.11) gives $M - M_0 = O(\sqrt{\mu})$; its numerator contribution is smaller. The mean bound is $O(\mu^2/n^2)$. This proves (9.9) for all competitors, including those initially in extreme or rare parity regimes. $\square$

Proof of Theorem 9.1. The explicit tuned law gives the required upper bound by Theorem 13.1 and Lemma 13.2. For each n , choose a competitor within $o(\varepsilon_n)$ of the infimum. Such an approximate minimizer exists by the definition of infimum; attainment is unnecessary. The upper bound makes it $O(\varepsilon_n)$ competitive, so Proposition 14.3 applies. Its coordinates

$$\alpha_n = \frac{(\nu_n - \nu_0)n^2}{\gamma\mu^2}, \quad \beta_n = \frac{9(M_n - M_0)}{5\sqrt{\mu}}$$

are bounded. These definitions reproduce the integer M_n exactly in (13.1); no rounding ambiguity remains. Every subsequence has a convergent further parameter subsequence. The uniform local theorem on that subsequence gives a limiting scaled error at least $\frac{\gamma}{2} \min J = \gamma \log 2\sqrt{2/\pi}$. This proves the global lower bound and hence (9.8).

For any asymptotically optimal sequence, the same boundedness and compactness hold. Uniqueness of the minimizer in Lemma 13.2 forces every parameter limit to be $(2 \log 2 - 3, 0)$, yielding (9.10). Conversely those relations put the parameters in a compact local set and force exactly that limit, so Theorem 13.1 proves asymptotic optimality. All conclusions are uniform in the two parities. $\square$

15 Classical context and the precise scope of the result

15.1 What the comparison adds

Positive binomial fitting, Charlier corrections, and Hermite absolute moment limits are established approximation mechanisms. Soon [9] chooses an integer trial count near a mean squared divided by a second factorial moment quantity, then preserves the mean. Barbour and Xia [8] discuss positive binomial fitting to a two cumulant reference; their Theorem 4.1, equation (4.2), already displays an error mechanism of size $q^2/\sqrt{Mq}$ in the small success probability regime. Čekanavičius, Peköz, Röllin, and Shwartz [10] construct shifted binomial fits and explicitly treat rounding and support effects. We do not claim these general constructions as new.

The rate $n^{-5/4}$ also has a classical cumulant explanation. If $f_2 = \kappa_2 - \kappa_1$ and $f_3 = \kappa_3 - 3\kappa_2 + 2\kappa_1$ are factorial cumulants, Theorem 11.2 gives

$$f_2 = -\frac{3}{2}c^2 + O(t), \quad f_3 = \frac{33}{4}c^3t + O(t^2).$$

A binomial matched in mean and variance before rounding has $f_{3,B} = 2(\nu - v)^2/\nu = \frac{9}{2}c^3t + O(t^2)$. The leading discrepancy is $\frac{15}{4}c^3t$, which upon division by $3!\mu^3$ suggests the Charlier coefficient $5/(8n^2)$. This calculation predicts the scale and coefficient. It is not a proof of the TV equivalent: it does not exclude another rearrangement of probability mass at that scale. The weighted estimate in Theorem 12.1 supplies exactly that missing distributional control, together with both support tails.

Barbour, Kowalski, and Nikeghbali [7] give general mod discrete and Poisson–Charlier expansion methods, including positive approximants. Chhaibi, Delbaen, Méliot, and Nikeghbali [11], Theorem 3.11, give sharp total variation asymptotics governed by Gaussian derivatives and their absolute integrals under suitable global residue hypotheses. Méliot, Nikeghbali, and Visentin [12] develop higher order mod Poisson approximation schemes and a positive conversion. These are important general precedents for the final Hermite step. The short weighted median minimization in Lemma 13.2 is elementary L^1 projection, not a new general optimization principle.

There are specific reasons not to insert this law unchanged into an off the shelf ordinary Poisson or independent Bernoulli theorem. Its characteristic function has modulus one at frequency π , while an ordinary Poisson characteristic function there has modulus $e^{-2\mu}$. Thus its ordinary Poisson deconvolution residue is not uniformly bounded over the full Fourier interval. Also, a nondegenerate ordinary sum of independent Bernoulli variables has consecutive support after deterministic shifts; R_n lives on one parity class. Reindexing the lattice or proving a conditioning transfer would require additional work. The common conditioned reference here resolves this directly.

There is a second precision issue: the ordinary Poisson residual corresponding to $\gamma C_3/n^2$ has cubic coefficient $\gamma\mu^3/n^2$, which tends to zero. An unscaled sharp theorem with a limiting residual coefficient does not by itself yield a relative equivalent at this smaller scale. One must verify a suitably normalized, quantitative remainder. Finally, the full binomial infimum requires the global parameter localization of Section 14. It is not a consequence of three cumulants or a local fitting calculation.

The article’s asserted result is therefore the model specific weighted comparison, explicit coefficient and tuning, and global localization for all defined parity conditioned binomials. We make no worldwide priority claim and no claim of external peer review. The analytic proof is independent of the computational checks.

15.2 OEIS connection and source limits

The indexed primary entry A360592 [1] records (9.1), the finite sum, and initial terms

$$1, 1, 2, 5, 14, 44, 149, 543, 2096, 8539, 36444, 162380, \dots$$

Its posted leading equivalent is $(n/2)^{n/2} e^{c\sqrt{n}-3c^2/4}/2$. The scalar result in Report 230, reproduced in Proposition 11.1, establishes this and gives first relative coefficient $c/4 + 11c^3/8$. The indexed entry's displayed intended refinement contains an additional $1/(8c)$. As a claim about that first correction it disagrees with the proved coefficient; interpreted only as a leading equivalence under $\sim$, the displayed formula remains true. This article does not depend on correcting the online entry and makes no assertion about the origin of that extra term.

Primary sources were checked on 5 October 2026. The OEIS direct fetch was unavailable, so the entry information comes from its indexed primary page; its latest revision history was not independently verified. The bibliography identifies inspected primary documents and versions. Ehm's 1991 publisher abstract [15] was available, but its full original theorem was not inspected; no theorem level assertion from it is used. The named general papers above provide context, not black box proof inputs. A bounded source comparison cannot certify historical absence of a related exact constant or optimizer.

[write] **7 October 2026.** The write read A360592 directly: revision #28 of 17 February 2023, the latest, with the finite sum, the initial terms displayed here and the leading equivalent as stated, and with the refinement whose first correction carries the extra $1/(8c)$ (Remark 7.1).

15.3 Further research directions

The following directions are outside the present theorem.

1. **Next order optimization and integer effects.** The theorem leaves bounded trial changes undetermined. A further weighted expansion could identify how the fractional part of the real trial count enters the next TV term, and whether it selects particular rounding choices. It would still require a separate argument to identify exact finite n minimizers.
2. **Larger positive families.** An additional even translation, a mixture, or another positive family may remove part of the cubic residual. Positivity, support, and the dimension of the limiting L^1 projection would need to be verified for each family.
3. **Uniform parameter and modulus regimes.** Other fixed parameters in the self powered sum family lead to conditioning modulo more than two. Such extensions are outside this article. Uniform estimates when the modulus or model parameter grows with n require substantially different root and tail control; no such uniformity follows from a fixed parameter expansion.
4. **Effective error certificates.** The proof gives asymptotic O estimates with unspecified constants. Explicit onset thresholds and interval arithmetic could turn the positive construction into a certified finite n error bound, especially where the moment matched and tuned errors cross.

5. **General deformed lattice laws.** It is useful to isolate hypotheses on a positive deformation that imply a weighted Charlier remainder together with a global two parameter localization principle. Such a theorem would need tail control and identifiability, not just a list of cumulants.
6. **Beyond fixed order.** The marked generator works at every fixed order. It does not show convergence as the order grows, justify optimal truncation, or determine an exponentially small parity correction. Those questions need additional analysis.

[write] **Added 7 October 2026, under Vladimir's standing rule of 4 October 2026.** Every open question of this Part is in this list. Its third item meets Part I, which treats every fixed $p \geq 2$ with residue conditioning modulo $p + 1$ but proves only Poisson references there; the binomial optimization for $p \geq 2$ is open. No claim of the source was found false.

16 Reproducible computation and its limits

The accompanying source archive contains this full article, its PDF, public Python programs, dependency versions, immutable check receipts, and a manifest. The proofs above stand independently of the programs. This section states what the reproducible checks do and what they cannot certify.

16.1 Exact finite checks

The exact rational implementation constructs the defect polynomials, scalar coefficients, logarithmic marked coefficients, and tilted signed density polynomials. It checks the cubic logarithmic difference (12.18), its identification with (12.19), and the algebraic characteristic function derivative identities (14.7). Small integer coefficient checks compare the finite sum with the coefficient construction from (9.1); marked weights and their recurrence are also checked. These finite algebraic identities are exactly checked, whereas the unbounded order remainder claims are proved in the article.

The supplementary residue coupling is also checked with exact rational weights on a bounded deterministic collection of positive log concave laws, translated supports, and moduli $2, \dots, 7$. The tests compare all present residue pairs and all quantile constancy cells, including boundary values. These are finite examples supplementing the proof in Appendix A, not a computational proof for arbitrary log concave laws.

For numerical computation, the exact adjacent marked weight recurrence is particularly convenient. With $m = (n + r)/2$ and $k = (n - r)/2$,

$$\frac{W_n(r + 2)}{W_n(r)} = \frac{k(1 + 1/m)^k}{(r + 1)(r + 2)}. \quad (16.1)$$

Indeed m increases by one and k decreases by one, so division of the factorial formulas in (9.2) gives this identity immediately. The program anchors at a central mass and propagates the relative weights. Reference Poisson and binomial masses have separate elementary recurrences. Normalizing the exact weights in a central window avoids summing through all n lattice points.

16.2 Tail formulas and numerical evidence

For an ordinary Poisson or binomial law of mean η , exponential Markov bounds outside a retained interval $[\ell, h]$ use

$$\exp\{-\eta + x - x \log(x/\eta)\}$$

at the applicable lower and upper endpoints x , provided the endpoint is on the corresponding side of the mean. Divide by the exact parity probability for each conditioned reference. For the exact law use (10.3): its omitted unnormalized mass relative to $Q_{\lambda,a}$ is at most $e^{1/(4n)}$ times that reference tail. Dividing by the retained positive normalizer gives an upper bound for the omitted exact probability. If its omitted probability is at most τ and the reference omitted probability is at most ρ , the half ℓ^1 discrepancy computed with the normalized exact window masses and unnormalized reference window masses differs from the true TV by at most $\tau + \rho/2$. The two contributions are exact window renormalization and the discarded tail sum.

The formulas for these bounds are analytic. Their saved decimal evaluations and the computed probabilities use high precision floating point arithmetic, not directed interval arithmetic. In particular, a displayed extremely small tail value is not a bound on floating point roundoff. Numerical results are corroboration, not effective rigorous error certificates.

The default numerical receipt uses 80 decimal digits and both parities near 10^4 , 10^6 , and 10^8 , with the variance polynomial v_+ through t^2 . The following are ratios to the corresponding proved sharp equivalents.

n	Moment matched ratio	TV tuned ratio
10^4	0.709741899	1.343673290
$10^4 + 1$	0.706720977	1.329789806
10^6	0.961167258	0.997326441
$10^6 + 1$	0.961020163	0.997786129
10^8	0.996103638	0.999555465
$10^8 + 1$	0.996032957	0.999454510

The tuned construction need not improve the finite n error at every tested value; the first rows illustrate why the theorem is asymptotic. The weighted $O(n^{-3/2})$ remainder is smaller than the leading $n^{-5/4}$ error only by a relative $O(n^{-1/4})$ factor. The receipts also record the simpler hierarchy, normalized density remainders, and evaluated tail bounds. Optional bounded command line runs allow larger n ; they do not change frozen fixtures.

[write] The table recomputed, 7 October 2026. The write recomputed the twelve ratios, and the four of the simpler hierarchy, with its own code (exact law from log-gamma values over $|r - \lambda| \leq 40\sqrt{\lambda}$, conditioned references by direct masses, 50 digits; M_0 by rounding half up). All agree with the table and with `data/235-binomial-numerical_receipt.json` in every printed digit. The printed values are roundings; the truncations that differ are 0.709741898, 0.706720976, 1.329789805, 0.961167257, 0.997326440, 0.996103637, 0.999555464 and 0.996032956. The constants of Section 9 are 0.3456571460..., 0.4718790625... and the ratio 0.7325121488... (printed as roundings, “about 0.34566”, 0.47188, “about 0.73251”).

The rare parity check enumerates 205 defined cases over a finite grid of trial counts and success probabilities, including endpoint and very small parity probabilities. It verifies the signed moment identities and the mean and variance inequalities. Six characteristic function checks compare the

exact derivative formulas with high precision differentiation and examine the least singular value and an ε_n scale displacement. These finite tests do not replace the uniform proofs of Lemma 14.1 and Proposition 14.3.

16.3 Build and verification contract

The archive README gives complete commands. The public build verifies every manifest entry and the full source inventory, runs exact, numerical, and guard checks without updating receipts, and compiles a disposable copy of the TeX sources. Shell escape is disabled both for the private TeX format and for document compilation. Python child processes use the no bytecode flag; the integer string conversion cap is 640 digits. Preconditions use explicit exceptions, so optimized Python does not remove guards.

Output paths must be new, outside the source tree, and free of symlink ancestors and traversal components. No existing output is overwritten. The ZIP is assembled deterministically with fixed member order, timestamps, modes, and stored compression. The actual archive replay extracts independently for normal and optimized Python, compares every member's bytes and metadata, rebuilds the PDF and receipts, and requires equality of the final ZIP bytes and unchanged source trees. This is byte reproducibility on the recorded toolchain, not a promise that a different TeX or symbolic algebra version will produce identical bytes. A manifest is an integrity record, not a digital signature or an independent proof of correctness.

[write] The shipped files, 7 October 2026. In ProveIt the PDF, `MANIFEST.sha256`, the delivered README and the delivered TeX sources are not shipped (this Part is the text); the programs are `code/235-binomial-*.py`, the receipts and requirements `data/235-binomial-*`. The builder and the archive replay expect the delivered layout; the report README gives commands that rerun the checks in a scratch copy.

[write] Independent check of the write, 7 October 2026. An independent adversarial check of the write read the three OEIS entries again (A360592 #28, A360479 #40, A360747 #22, unchanged since February 2023) and recomputed every number the write added, with its own code and, where one exists, by a different method.

- *The refutation of the OEIS first corrections* (Remark 7.1). The check typed the three formula lines afresh from the live entries and used them directly, not the source's transcription; SymPy confirms that their leading factors are $M_1(n)$, $M_2(n)$, $M_3(n)$ and that their multipliers are those displayed in Section 7.4. It evaluated $a_p(n)/M_p(n)$ at n up to 10^{24} by a route the write did not use: one central summand by log-gamma, all others by the exact adjacent ratio of consecutive summands (agreeing with the exact integers at $n = 10^4$ to 10^{-56}). Then (entry $-a_p(n))/(M_p(n)t$) is 0.109664, 0.107465, 0.107244, 0.1072229, 0.10722073 at $n = 10^6, 10^8, \dots, 10^{14}$ ($p = 1$), 0.046842, 0.047452, 0.0475147, 0.04752092 at $n = 10^9, \dots, 10^{18}$ ($p = 2$), and 0.018632110, 0.018632475, 0.018632164, 0.01863213 at $n = 10^{12}, \dots, 10^{24}$ ($p = 3$), converging to $1/(8c_p) = 0.10722048562\dots, 0.04752160739\dots, 0.01863212660\dots$. The leading equivalents are right and the first corrections are wrong, as stated.
- *Coefficients.* An own generator, which expands the exact defect (2.4) by SymPy series (not through (3.6)) and averages by Touchard polynomials, gives every printed $C_{2,1..4}$ and $C_{3,1..6}$, $U_1 = -A\lambda^2 - b\lambda$ for $p = 2, \dots, 5$, $C_{p,1} = \dots = C_{p,p-2} = 0$ for $p = 4, 5$ with first nonzero coefficients $-81c^2/10$ and $-151c^2/12$, and from its own $C_{p,k}$ and U_j ($k, j \leq 12$) Table 1 and its truncations. Against the evaluations above, the residual after the printed coefficients, divided by the next power of t , is stable (24.192 for $p = 1$ after $C_{1,4}$, the value of the write's $C_{1,5}$; 12876.7 for $p = 2$ after $C_{2,4}$), and with the write's $C_{1,5}, C_{1,6}$ the order- t^7 ratio is $-5.50, -10.74, -11.30, -11.35, -11.36$ at $n = 10^4, \dots, 10^{12}$.

- *Decimals.* The critical values at $n = 10^4$ and $10^4 + 1$, the four marked ratios at $n = 10^9$ (exact law by the adjacent ratio, conditioning by the root-of-unity filter), the twelve ratios of the table of this section (exact law by (16.1) from a log-gamma anchor) and the constants $0.3456571460\dots$, $0.4718790625\dots$, $0.7325121488\dots$ agree with the write's notes in every digit, and every truncation stated in those notes is correct.

It also confirmed the provenance figures (archive sizes, file and line counts, 30 and 23 delivered pages, the two manifests), the byte identity of all 23 staged code, data and guide files, the word counts of the three passages printed once, the label numbering (229 delivered labels unchanged up to the shift of eight, 234 in all), Remark 6.7 against the volume's statements, and the README listing. No mathematical claim of the write was found wrong. One build defect was: the Guide's reading-conventions table and Table 1 both received the PDF destination `table.1`, so links to Table 1 led to the Guide (the write's log had two "duplicate ignored" warnings that its README reported as absent). The check gave the Guide's table its own anchor name; no number changed.

A A bounded residue coupling for log concave weights

This appendix gives a general supplementary lemma. It is not needed for the parity specific proof above, but clarifies why very rare residue conditioning can still preserve coarse location and scale. Its ingredients are classical log concavity and monotone likelihood ratio order; see [13, 14]. We give the full modular argument, without a novelty claim for this formulation.

Theorem A.1 (Bounded coupling of all present residues). *Let $w_0, \dots, w_N$ be strictly positive and log concave, meaning $w_k^2 \geq w_{k-1}w_{k+1}$ whenever $1 \leq k < N$. Normalize them to a probability mass function X . Fix an integer $q \geq 2$, and let X_a have its conditional law on residue a modulo q whenever that residue is present. There is a simultaneous coupling with*

$$|X_a - X_b| \leq q - 1 \quad \text{almost surely for every present pair } a, b.$$

For each fixed present residue there is also a coupling with the original law such that $|X_a - X| \leq q - 1$ almost surely. The common quantile coupling has this property. The statement extends to any translated consecutive finite support.

Proof. We first record a cross product proof of the required stochastic order, including zero endpoints. If probability masses P, Q on the integers satisfy

$$P(i)Q(j) \geq Q(i)P(j) \quad (i < j),$$

then for every integer t ,

$$F_P(t) - F_Q(t) = \sum_{i \leq t < j} \{P(i)Q(j) - Q(i)P(j)\} \geq 0.$$

Thus P is stochastically no larger than Q . A decreasing likelihood ratio on overlapping interval supports implies the cross product condition when the numerator support begins and ends no later than the denominator support. This formulation avoids division by zero.

For present residues $0 \leq a < b \leq q - 1$, write $X_a = a + qK_a$, $X_b = b + qK_b$ and $A_a = \lfloor (N - a)/q \rfloor$, $A_b = \lfloor (N - b)/q \rfloor$. The two index supports are $[0, A_a]$, $[0, A_b]$, with $A_a - 1 \leq A_b \leq A_a$. Log

concavity makes $r_j = w_{j+1}/w_j$ nonincreasing. Therefore for each positive shift d , the ratio $w_{t+d}/w_t = \prod_{j=0}^{d-1} r_{t+j}$ is nonincreasing wherever defined.

On the common support the mass ratio of K_b to K_a is a positive constant times w_{b+qk}/w_{a+qk} , hence decreasing. Both supports start at zero, and the numerator ends no later. The cross product criterion gives $K_b \leq_{\text{st}} K_a$.

Next compare $K_a - 1$, supported on $[-1, A_a - 1]$, with K_b , supported on $[0, A_b]$. The former begins and ends no later. On their overlap its likelihood ratio is a constant times

$$\frac{w_{a+q(k+1)}}{w_{b+qk}},$$

whose shift $q + a - b$ lies in $\{1, \dots, q - 1\}$; this ratio is decreasing too. At the left endpoint only the first distribution may have mass, and at a possible unmatched right endpoint only the second may have mass. Thus all endpoint cross products have the same orientation, and

$$K_a - 1 \leq_{\text{st}} K_b \leq_{\text{st}} K_a. \quad (\text{A.1})$$

This includes singleton supports and empty overlaps. Residues with no support are simply omitted.

Let $Q_a(u)$ be the lower quantile of K_a for $0 < u < 1$. Stochastic order is equivalent to pointwise quantile order, so (A.1) gives $Q_a(u) - 1 \leq Q_b(u) \leq Q_a(u)$. With a single uniform random variable U , set $X_a = a + qQ_a(U)$ for every present residue. For $a < b$, their difference is either $b - a$ or $b - a - q$, both of absolute value at most $q - 1$. This proves the simultaneous bound.

Choose independently a residue J with probabilities $\mathbb{P}(X \equiv j \pmod{q})$, and set $X = X_J$ on this coupling. Its law is the original mixture and it lies within $q - 1$ of every X_a . The ordinary common quantile coupling also works: at each u , the mixture quantile lies between the smallest and largest component quantiles. Below the smallest all component distribution functions are less than u , and at the largest all are at least u . These extreme quantiles differ by at most $q - 1$, proving the assertion. Translation changes only residue labels and leaves all distances unchanged. $\square$

Corollary A.2 (Coarse binomial bounds for any fixed modulus). *For any defined residue conditional of $X \sim \text{Bin}(M, \pi)$, with ordinary mean ν and variance v ,*

$$|\mathbb{E}X_a - \nu| \leq q - 1, \quad \text{Var } X_a \leq 2v + 2(q - 1)^2.$$

For every interval I of length L , there is a constant C_q such that

$$\mathbb{P}(X_a \in I) \leq C_q(L + 1)/\sqrt{v + 1}.$$

If q is fixed and $v \rightarrow \infty$, then $(X_a - \nu)/\sqrt{v} \Rightarrow N(0, 1)$ for any sequence of present residues.

Proof. Interior binomial weights are positive and log concave because $w_{k+1}/w_k = \pi(M - k)/((1 - \pi)(k + 1))$ decreases in k . Endpoint probabilities give point masses and need no conditioning argument. In the bounded coupling, the mean bound follows immediately and Minkowski's inequality gives $\mathbb{E}(X_a - \nu)^2 \leq (\sqrt{v} + q - 1)^2 \leq 2v + 2(q - 1)^2$. Variance is no larger. If $X_a \in I$, the coupled X lies in the interval expanded by $q - 1$ at each end. The ordinary Fourier atom bound proved in Lemma 14.1 bounds its mass by $C_q(L + 1)/\sqrt{v + 1}$. Finally the standardized coupling displacement is at most $(q - 1)/\sqrt{v}$, so the ordinary binomial characteristic function central limit argument transfers the limit. $\square$

For $q = 2$, these estimates give an alternative to the first part of Section 14, with no conditioning probability in any denominator. They do not supply the model specific cubic density expansion. Consequently this appendix makes no global approximation claim for another endpoint parameter or growing modulus. The uniform coupling distance $q - 1$ itself cannot be reduced: take $N = q - 1$ and a binomial success probability tending to zero. Its residue $q - 1$ conditional is concentrated at $q - 1$, while the unconditional mean tends to zero.

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